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# Navigating towards efuel: A scientometric insight into the application of membrane reactors

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ABSTRACT

The urgent need to decarbonize high-emission sectors has driven the development of Power-to-X technologies, which convert renewable electricity into electrofuels (efuels). Despite their potential, efuel production faces challenges such as high energy demand and low conversion efficiency. Membrane reactors, which integrate reaction and separation, offer a promising solution by improving yields and reducing energy requirements. This review presents a scientometric analysis of membrane reactors for efuel production using the Scopus database from 2003 to 2024. Analyzing 30 publications, six thematic clusters were identified using VOSviewer and Bibliometrix. Keyword co-occurrence and factorial analyses highlight main research themes and emerging areas, revealing gaps in reactor configuration optimization. Influential studies show that membrane reactors can enhance $CO_2$ conversion and methane yield compared to conventional systems, though challenges remain in membrane selectivity, economic viability, and long-term durability under real feedstock conditions. Additional issues include scalable module manufacturing and the lack of harmonized techno-economic, life cycle, and performance metrics. Sector-specific analysis identifies positive dynamics, such as compatibility with existing infrastructure, improved energy security, and supportive policies, as well as negative dynamics, including high production costs, resource competition, technological uncertainties, and new safety and regulatory requirements. By mapping research progress, this study provides insights to guide the advancement of membrane reactors and support sustainable efuel production and decarbonization goals.

## 1. Introduction

The global energy landscape is changing rapidly due to the need to address environmental damage and ensure a steady supply of energy for a growing population. Rising greenhouse gas emissions, mainly from burning fossil fuels, have made climate-related issues worse, such as rising sea levels, ecosystem disruption, and extreme weather events. International agreements, like the 2015 Paris Agreement, emphasize the importance of limiting global warming by shifting to low-carbon energy systems. Many countries have increased their renewable energy capacity, and projections show a significant increase in clean energy contributions over the next decade [1]. In 2024 alone, a record 585 GW of renewable power capacity was added globally, meaning renewables accounted for over 90% of all new power capacity expansions during the year, by solar and wind energy. The International Energy Agency estimates that renewables' share of global electricity generation could rise from about 32% in 2024 to roughly 43% by 2030. Despite this momentum, significant regional disparities remain, and the current pace of growth is still insufficient to meet the international goal of tripling renewable capacity by 2030 [2,3]. While renewable sources are crucial for reducing dependence on fossil fuels, their inconsistent availability poses challenges for maintaining grid stability and ensuring a constant energy supply. This calls for advanced technologies in energy storage and conversion [4].

A key sector contributing to emissions is transportation, which in 2022 accounted for 21.6% of global carbon dioxide ($CO_2$) emissions

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(7.95 Gt), with over 90% of its energy derived from oil-based liquid fuels. The sector's slow adoption of renewable energy, coupled with its reliance on existing infrastructure, underscores the need for sustainable fuel alternatives that can integrate seamlessly without requiring extensive infrastructural changes. Power-to-X (PtX) technologies can convert renewable electricity into chemical energy carriers, such as electrofuels (efuels) and biofuels, which can decarbonize transportation while using current systems. Research identifies three primary strategies for this transition: (i) direct electrification through battery-powered vehicles, (ii) hydrogen-powered vehicles using indirect electrification, and (iii) non-fossil fuels like efuels for conventional internal combustion engines [5,6].

Green hydrogen, made through electrolysis powered by renewables, is at the center of PtX. It can be used directly or turned into synthetic chemicals. However, global electrolysis capacity, while growing fast, still faces limitations due to hydrogen's low energy density, storage problems, and safety risks like corrosion and leaks [7]. This storage requirement is necessary due to the intermittency of renewables. For instance, a 100 MW solar facility may generate a significant energy surplus (30–50% above immediate demand), which must be stored as hydrogen to ensure electrolyzer utilization and economic viability [8]. The resulting hydrogen must then be stored for durations ranging from hours for grid balancing to months for seasonal energy shifting, each application imposing distinct technical demands [8]. Common storage methods face a critical trilemma between energy density, efficiency, and cost. For example, while liquid hydrogen offers high density (~8.5–10 MJ/L), its round-trip efficiency is only 55–65% due to energy-intensive liquefaction [9]. It is these inherent limitations that create a compelling rationale for converting hydrogen into chemically stable carriers like e-ammonia or e-methanol. These efuels offer higher volumetric energy densities, can use existing global infrastructure, and often present a net efficiency advantage when conversion losses are compared to the combined storage and transport losses of pure hydrogen [10]. For example, even when compressed to 700 bar, hydrogen's volumetric energy density (~5.6 MJ/L) remains an order of magnitude lower than that of liquid fuels like gasoline or diesel (see Table A1). To solve these issues, efuels such as e-methane, e-kerosene, e-ammonia, and e-methanol are getting more attention. They store more energy, work with today's infrastructure, and are easier to transport. These fuels come from green hydrogen and capture $CO_2$, giving a way to cut emissions in sectors that are slow to switch to electricity, like aviation, shipping, and long-distance road transport. Unlike biofuels, which need biomass, efuels are more flexible. They can reduce emissions and use existing infrastructure while pairing renewable electricity with $CO_2$ capture [11].

The production of efuels, however, faces several technical and environmental barriers. Traditional synthesis processes, often reliant on $H_2$ derived from fossil-based methane ($CH_4$) steam reforming, generate considerable $CO_2$ emissions, undermining their sustainability. For example, synthesizing ammonia requires high-pressure and high-temperature conditions (typically 400-500°C and 100–200 bar) using iron or ruthenium catalysts, while methanol production operates at 200-300°C and 50–100 bar with copper-based catalysts [12]. Fischer-Tropsch synthesis, used for producing hydrocarbons, i.e., e-kerosene, operates under similarly demanding conditions, with low conversion efficiency and challenges in utilizing $CO_2$ directly due to catalyst deactivation from water by-products [13]. Hydrogen recovery from ammonia decomposition or methanol reforming further complicates the process, requiring high temperatures (e.g., 700°C for ammonia cracking) and energy-intensive purification methods such as pressure swing adsorption [14]. These inefficiencies highlight the need for advanced technologies that optimize reaction conditions, reduce energy consumption, and enhance scalability for sustainable efuel production.

Membrane reactors are an important technology for efuel synthesis. They combine reaction and separation in one unit and address key limits of conventional processes. First, they shift reaction equilibrium by selectively removing products such as water, which increases single-pass conversion beyond thermodynamic limits [15]. Second, they improve selectivity and protect catalysts by controlling reactant supply or removing inhibitors, which stabilizes yields and extends catalyst lifetime [16]. Third, they intensify the process by combining several unit operations into one system. This reduces energy use, equipment size, and system complexity [17]. These features make membrane reactors suitable for efficient and scalable efuel production in PtX systems. In practice, selective membranes embedded in catalyst-filled reactors enable product removal through pressure-driven transport. This allows higher yields under milder conditions and reduces side reactions and catalyst deactivation. Membrane reactors can also function as reactant distributors or as contactors for selective product extraction [18]. These features make them suitable for compact and flexible applications, including maritime systems and off-grid power. Despite these advantages, technical challenges still limit large-scale deployment, and continued research is needed to support industrial adoption and reduce reliance on fossil fuels.

Since the topic of efuel production is currently of significant relevance, as evidenced by the themes of research, several studies have been conducted based on membrane reactor technologies and efuel production processes [6,19,20]. Yet their combined application remains underexplored, with no prior studies utilizing scientometric analysis to evaluate the progress of research in this integrated field. As a result, this paper presents the following contributions:

- Conducts a scientometric assessment (2003–2024) to map the research landscape, identifying seminal publications and tracing the evolution of membrane reactor applications in efuel production.
- Characterizes international collaborative networks and delineates emerging methodological approaches through keyword co-occurrence and thematic mapping.
- Provides a sector-specific critical analysis that outlines key positive and negative dynamics, including infrastructure, production costs, and policy support, with practical insights for shifting from conventional fuels to sustainable efuels.

To guide the systematic assessment of research progress and thematic evolution, this study addresses the following research questions (RQs):

**RQ1:** What are the main publication trends over the past two decades, including growth over time, international collaboration, and leading journals?

**RQ2:** What are the main research themes and topic clusters in this field based on keyword analysis and thematic mapping?

**RQ3:** Which articles, authors, and countries have had the greatest impact on this research area, and how do they collaborate?

**RQ4:** What membrane reactor technologies are used to produce different efuels, and what are their performance results and challenges?

**RQ5:** What are the key opportunities and barriers in moving from conventional fuels to efuels, and what practical lessons can industry and policymakers learn?

This paper is structured into six sections: Section 2 details the types of efuels; Section 3 describes the methodology; Section 4 presents the results and analysis; Section 5 provides a discussion, and Section 6 concludes the study.

## 2. Types of efuel

The diverse spectrum of efuels presents promising opportunities for advancing sustainable energy systems, particularly within sectors that are challenging to electrify. Among the various efuels, e-methane, e-methanol, e-ammonia, and e-diesel stand out due to their potential scalability, versatility, and ability to integrate into existing infrastructure. Each of these fuels offers unique production pathways, applications, and challenges. This section explores these efuels in detail.

### 2.1. E-methane

E-methane, a synthetic form of $CH_4$ produced through the methanation process, is an alternative to natural gas. This compound appears when $H_2$, ideally sourced from renewable energy systems, reacts with $CO_2$ in a controlled chemical reaction. The process begins with electrolysis, where water is split into $H_2$ and oxygen ($O_2$) using renewable electricity, followed by the introduction of this $H_2$ into a methanation reactor. Here, it combines $CO_2$ at temperatures ranging from 250 to 400°C and pressures of 5 to 50 bar, following the Sabatier reaction [21].

$$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$$

This method requires approximately 5.5 kg of $CO_2$ for every kg of $H_2$, a demand that often necessitates transportation from carbon capture and storage facilities, thereby increasing logistical costs and complexity [22]. The resulting e-methane can be integrated into existing natural gas grids that will provide a versatile fuel for heating, electricity generation, and powering natural gas vehicles. A prime example of this technology in action is the Audi e-gas project in Germany, recognized as the world's largest power-to-gas facility. Operational since 2013, it produces about 1000 tons of synthetic $CH_4$ annually, utilizing $CO_2$ from a biogas plant and distributing the output through established infrastructure to compressed natural gas refueling stations [23]. This initiative not only demonstrates the feasibility of storing renewable energy in chemical form but also shows e-methane's potential to mitigate greenhouse gas emissions when production relies entirely on renewable power sources.

### 2.2. E-methanol

E-methanol ($CH_3OH$) is a versatile alcohol with potential as both a fuel and a building block for industrial chemical production, traditionally derived from fossil resources such as natural gas and coal. Global production currently stands at 98 million tons per year, releasing approximately 0.3 gigatons of $CO_2$ annually, which constitutes about 10% of the chemical sector's emissions. Forecasts predict a surge in demand to 500 million tons by 2050, potentially escalating emissions to 1.5 gigatons yearly if fossil fuel dependency continues unabated [24]. In contrast, e-methanol offers a sustainable alternative, synthesized from renewable feedstocks like biomass or captured $CO_2$, paired with green hydrogen generated through renewable energy sources [25]. The production process relies on catalytic hydrogenation, where $CO_2$ and $H_2$ in a 1:3 ratio react within a reactor at 200–300°C and 50–100 bar, employing Cu/Zn/Al-based catalysts to facilitate the reaction [26].

$$CO_2 + 3H_2 \rightarrow CH_3OH + H_2O$$

An emerging technique involves converting e-methane directly into e-methanol using water as an oxidizing agent and a copper-zeolite catalyst, though this method remains experimental, with industrial application likely 5 to 20 years away [27]. E-methanol's energy density, roughly half that of gasoline, suits it for use in modified spark-ignition engines, while its chemical properties support the manufacture of products like formaldehyde and dimethyl ether (DME). China leads global adoption, establishing standards for M15 and M85 fuel blends containing 15% and 85% methanol, respectively, and pilot projects such as the Mef$CO_2$ plant in Germany, which produces 500 tons annually using $CO_2$ from coal-fired plants, illustrate its practical implementation [28]. This dual role positions e-methanol as a key player in decarbonizing transportation and industrial processes.

### 2.3. E-ammonia

E-ammonia is a compelling option both as an alternative fuel and as a medium for transporting $H_2$, with approximately 80% of its conventional production dedicated to agricultural fertilizers [28]. The remaining output supports the synthesis of materials such as plastics, synthetic fibers, and explosives. Crucially, about 98% of global ammonia is currently derived from fossil fuels like natural gas and coal, a process that contributes around 1.8% to worldwide $CO_2$ emissions [29,30]. The traditional Haber-Bosch process drives this production, combining nitrogen and $H_2$ at temperatures of 400–450°C and pressures of 150–200 bar, utilizing an iron-based catalyst to achieve a conversion rate of 25–35% per cycle.

In contrast, e-ammonia is produced using green hydrogen generated from renewable energy sources, offering a carbon-free alternative. This shift, however, introduces environmental considerations, including the risk of NOx emissions and potential ecological harm from ammonia leaks into soil, air, and water, which can lead to biodiversity loss and eutrophication [31]. As a fuel, e-ammonia powers gas turbines and internal combustion engines, often requiring thermal cracking at 400-600°C to manage its high ignition temperature, sometimes with a combustion promoter like diesel or $H_2$ derived from onboard reforming [32]. The Commonwealth Scientific and Industrial Research Organization in Australia has pioneered its use in fuel cells, equipping two specially designed vehicles with membrane technology that separates $H_2$ from ammonia for clean energy output [33]. Stored as a liquid at a modest pressure of 1 MPa, e-ammonia benefits from existing distribution networks, enhancing its practicality. A notable development is the Abu Dhabi National Oil Company's planned blue ammonia plant in Ruwais, United Arab Emirates, with a capacity of 1000 kilotons per year, underscoring its potential as a zero-carbon fuel through carbon capture and storage integration [34].

### 2.4. E-diesel

E-diesel is a synthetic diesel made using power-to-liquid technology, which helps create sustainable fuels that can be used with current infrastructure. This fuel is synthesized using the Fischer-Tropsch process, where $H_2$ and carbon monoxide (CO) react at temperatures between 190 and 350°C and pressures of 20 to 50 bar, catalyzed by iron or cobalt materials to form liquid hydrocarbons [35,36]. The process begins with the generation of syngas, either from $CO_2$ and $H_2$ via the reverse water–gas shift reaction or directly through co-electrolysis, followed by conversion into diesel-range hydrocarbons ($C_{12}$–$C_{22}$) [37]. Alternatively, e-diesel can be derived from methanol intermediates through the methanol-to-gasoline (MTG) pathway, involving sequential reactions starting with DME synthesis, olefin formation, and hydrotreating to refine the product [38]. The chemical reaction of DME synthesis is as follows:

$$2CH_3OH \rightarrow CH_3 - O - CH_3 + H_2O$$

The choice of catalyst is critical: iron-based catalysts, favored for their cost-effectiveness and flexibility with syngas ratios (0.5–2.5 $H_2$/CO), suit high-temperature operations, while cobalt-based catalysts, requiring lower temperatures (190–240°C) and a narrower ratio (2.0–2.3), enhance diesel yield [39]. E-diesel's chemical similarity to fossil diesel allows it to function as a drop-in fuel, requiring no engine modifications and offering the advantage of being sulfur-free, a trait that distinguishes it from biodiesel esters [40]. Companies such as Neste and Carbon Clean Solutions, in collaboration with LanzaTech, are driving its development, utilizing its compatibility with current refueling networks to support a transition to cleaner energy [41]. With an efficiency factor of 0.82 to 0.83, e-diesel's sustainability hinges on renewable raw material sources, positioning it as a promising solution for decarbonizing transportation.

Environmental impact and life-cycle performance are critical for evaluating e-diesel sustainability. Combustion of e-diesel releases $CO_2$, and climate benefits depend on a closed carbon loop based on renewable electricity and captured $CO_2$. Life-cycle assessments show strong sensitivity to the electricity source. Carbon intensity can be about 12 g$CO_2$e per MJ with renewable electricity, but can exceed 380 g$CO_2$e per MJ

with carbon-intensive grids, which may be higher than conventional diesel. E-diesel, therefore, offers climate benefits only when produced with low-carbon electricity and efficient process designs. Fischer–-Tropsch e-diesel can improve air quality due to high cetane number, low aromatics, and zero sulfur content, which reduces particulate matter and often lowers $NO_X$, CO, and hydrocarbon emissions, although tailpipe emissions are not eliminated. Power-to-liquid e-diesel has high energy demand, with well-to-wheels energy use roughly twice that of conventional diesel, which requires large renewable capacity. As a result, e-diesel is best suited as a transitional fuel for hard-to-electrify transport sectors under strict carbon and energy constraints [42,43].

The practical deployment of e-diesel extends beyond laboratory production and addresses real energy needs in heavy-duty sectors that rely on internal combustion engines. E-diesel can be used in existing gas stations, so major infrastructure changes are not required. This makes it suitable for near-term adoption and supports a transition from fossil fuels to renewable systems. Refining steps, such as fractional distillation, improve fuel quality by separating products based on boiling points. Key challenges remain. The Fischer–Tropsch process requires high energy input, and feedstock availability varies, which affects cost and scalability. Ongoing research on advanced catalysts, such as Fe–Mn–K systems, aims to improve conversion efficiency and selectivity toward longer hydrocarbon chains, which could enhance the environmental performance of e-diesel [44].

### 2.5. E-kerosene

E-kerosene is a synthetic aviation fuel made using power-to-liquid technologies, providing a practical solution to lower the carbon emissions from air travel, which contributes 2–3% of global emissions. This fuel is produced via the Fischer-Tropsch process, where $H_2$ and CO, derived from $CO_2$ and $H_2$ or syngas, react under controlled conditions of 190 to 350°C and 20 to 50 bar, guided by cobalt or iron catalysts to yield hydrocarbons in the $C_8$–$C_{16}$ range suitable for aviation [36]. For production, the process sequentially includes the compression of $H_2$ and $CO_2$, the generation of CO via the reverse water–gas shift, and then Fischer-Tropsch synthesis. After that, hydrocracking is performed to break long chains into kerosene-compatible molecules. This step enhances the quality, and as a result, its standard aligns with sustainable aviation fuel (SAF), allowing it to blend with conventional fuels like Jet A-1 or JP-8 without the need for aircraft modifications [45]. This process requires approximately 1.36 kg of $H_2$ and 14.55 kg of $CO_2$ per kg of SAF [25]. Leading companies such as SkyNRG, Neste, and LanzaTech are working on their development, while airlines like Lufthansa have successfully conducted test flights. As of 2021, Germany stands out as a key producer, aiming to expand usage to meet growing demand for cleaner aviation options [41].

The adoption of e-kerosene carries notable implications for the aviation industry's sustainability goals, offering a means to decarbonize a hard-to-electrify sector. Its compatibility with existing infrastructure and engines facilitates gradual integration, with test flights demonstrating its reliability, e.g., a Boeing 737 flight from Amsterdam to Madrid in 2021 used 500 L of e-kerosene blended with traditional fuel [46]. However, production remains limited, constrained by the complexity of feedstock composition and the small scale relative to conventional aviation fuel volumes [47]. Research continues to refine catalyst performance, with Fe–Mn–K catalysts showing promise for higher selectivity toward aviation-grade hydrocarbons. Economic feasibility hinges on $H_2$ costs ($0.8–$5 per kg) and $CO_2$ pricing ($0 to $76.2 per ton), with the minimum fuel selling price ranging from $4.30 to $6.78 per gallon [48].

### 2.6. E-gasoline

E-gasoline, another product of power-to-liquid technologies, is a synthetic alternative to conventional gasoline, addressing the need for renewable fuels in the transportation sector. It is synthesized through the Fischer-Tropsch process or the MTG route, where $H_2$ and CO react at 190 to 350°C and 20 to 50 bar, catalyzed by iron-based materials to produce hydrocarbons in the $C_4$–$C_{12}$ range [35]. The MTG process involves converting methanol into DME, followed by olefin synthesis and oligomerization, with hydrotreating to finalize the gasoline product [38]. Iron catalysts, operating at higher temperatures (320–350°C), are effective for this shorter-chain output, offering flexibility with syngas ratios (0.5–2.5 $H_2$/CO), though cobalt catalysts can also be used at lower temperatures (190–270°C) for refined selectivity. E-gasoline's chemical properties align with traditional gasoline, enabling its use in existing spark-ignition engines without modification, a key advantage for rapid adoption. Companies such as ExxonMobil and Haldor Topsoe, through the TIGAS process, have scaled this technology, with a 2011 Chinese plant achieving a 3000-ton daily capacity [49]. Its production efficiency, though lower than some efuels (0.82–0.83), depends on renewable $H_2$ and $CO_2$ availability.

Environmental impact and life-cycle performance of e-gasoline depend on the production pathway. E-gasoline can achieve large greenhouse gas reductions when renewable electricity and biogenic or atmospheric $CO_2$ are used. Under these conditions, carbon intensity can fall below 20 $gCO_2$-eq per MJ, which represents more than an 85% reduction compared to fossil gasoline. These benefits decrease quickly if electricity comes from carbon-intensive grids or if $CO_2$ is linked to fossil sources. Energy efficiency is a major challenge. Electrolysis operates at about 65% to 70% efficiency, and fuel synthesis operates at about 65% to 85% efficiency. Overall electricity-to-fuel efficiency ranges from 40% to 60%, which requires much more renewable electricity than direct electrification [50]. Large-scale deployment also stresses $CO_2$ supply chains. About 82 g of $CO_2$ is required per MJ of fuel, which increases energy use and infrastructure needs for capture, purification, and transport Combustion of e-gasoline can reduce some pollutants such as hydrocarbons and CO, but it does not eliminate $NO_x$ and particulate emissions [51]. In some cases, aromatic-related emissions may increase depending on fuel blend and engine design. Therefore, e-gasoline is best suited as a transitional drop-in fuel for existing vehicles, and its environmental benefit depends on low-carbon electricity, sustainable $CO_2$ sources, and use in sectors where electrification is difficult [52]. Table 1 provides an overview of key efuel production methods, applications, environmental considerations, and notable examples.

An important consideration for all these efuels is the carbon feedstock. Biogenic $CO_2$, which is a by-product from biogas plants or bioethanol fermentation, offers a practical and often lower-cost source. Utilizing this stream enables circular carbon recycling. However, it is crucial to distinguish this from carbon removal, which requires permanent storage of the biogenic carbon [50].

## 3. Methodology

This study employed a scientometric analysis to map research trends and identify themes at the intersection of membrane reactors and efuel production. Its reliance on quantitative methodologies helps to eliminate bias in systematic literature reviews. The bibliometric information was obtained from the Scopus database, which contains scholarly articles that aggregate pertinent research and offer unbiased summaries of findings. As one of the largest citation and abstract databases, it includes a wide range of peer-reviewed articles. It also provides useful tools for analysis, such as tracking citations, calculating the h-index, and identifying research trends. A systematic search method was applied on 16th August 2025 using Boolean operators to create specific queries. The OR operator was used to connect related keywords, AND linked different sets of keywords, and the NOT operator helped exclude unwanted terms. The query was set up as: (TITLE-ABS-KEY((“membrane reactor*” OR “membrane technology*” OR “membrane process*”) AND (“efuel production” OR “efuel processing” OR “electrofuels” OR “synthetic fuels” OR “power-to-liquid” OR “power-to-gas”)) AND NOT TITLE (review)).

**Table 1**
Summary of selected efuels: Production, application, environmental profile, and key challenges [21,43,106].

| Efuel | Production method and key reaction | Key applications | Environmental considerations | Key challenges | Examples |
|---|---|---|---|---|---|
| E-methane | Methanation (Sabatier): $CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$; Conditions: 250–400°C, 5–50 bar. Feedstock: $H_2$ from electrolysis, CO from capture. Requires approximately 5.5 kg $CO_2$ per kg $H_2$ input, corresponding to about 2.75 kg $CO_2$/kg $CH_4$ produced. | Integration into natural gas grids, fuel for vehicles, industrial energy, and electricity generation. | Lower carbon footprint with renewable $H_2$. Main risk: $CH_4$ leaks ($GWP_{100} = 28$). $CO_2$ is released upon combustion. | 1. High energy penalty from multi-step conversion (electricity → $H_2$ → $CH_4$).2. Climate performance is highly sensitive to methane leakage across the supply chain. 3. Requires low-cost clean electricity and low-carbon $CO_2$ sources.4. Catalyst durability concerns from impurities (sulfur, siloxanes). | Audi e-gas project (Germany), producing 1,000 tons/year. |
| E-ammonia | Haber-Bosch Process: $N_2 + 3H_2 \rightarrow 2NH_3$; Conditions: 400–450°C, 150–200 bar, iron catalyst (25–35% conversion/cycle). Feedstock: Green $H_2$ from renewables, $N_2$ from air separation. | Agricultural fertilizers (~80%), fuel for turbines/ engines, $H_2$ carrier via fuel cells or cracking. | Carbon-free if produced with green $H_2$. Main risks: $NO_X$ emissions from combustion; toxicity and ecological harm (eutrophication) from leaks. | 1. Lower volumetric energy density than hydrocarbons.2. Requires new infrastructure for bunkering, safety, and engine/ combustion systems.3. Maritime deployment hurdles: certification, crew training, port/community acceptance.4. Need to control $NO_x$ emissions and manage fuel slip. | CSIRO fuel cell vehicles (Australia); ADNOC's 1,000 kt/ year blue ammonia plant (UAE). |
| E-diesel | Fischer-Tropsch Synthesis: $(2n + 1)H_2 + nCO \rightarrow C_nH_{2n+2} + H_2O$Conditions: 190–350°C, 20–50 bar, Fe/Co catalysts. Alternate route: MTG pathway. | Drop-in fuel for existing diesel engines, heavy-duty transport (trucks, shipping). | Sulfur-free, chemically similar to fossil diesel. Sustainability depends entirely on renewable feedstock. Energy-intensive production. | 1. Competes with direct electrification on energy efficiency.2. High capital expenditure for PtL plants, challenging bankability.3. Logistical complexity of securing sufficient $CO_2$ supply.4. Managing intermittent renewable electricity supply to avoid cost escalation. | Neste, Carbon Clean Solutions with LanzaTech (development). |
| E-kerosene | Fischer-Tropsch with Hydrocracking: Fischer-Tropsch synthesis tailored for $C_8$–$C_{16}$ hydrocarbons, followed by hydrocracking. Conditions: 190–350°C, 20–50 bar.Inputs: Requires ~1.36 kg $H_2$ + ~14.55 kg $CO_2$ per kg SAF. | SAF for blending with conventional jet fuel (Jet A-1, JP-8). | Carbon-neutral potential with renewable inputs. Currently limited by small production scale and high cost (MFSP: ~$4.30–6.78/ gallon). | 1. Extremely high renewable electricity demand and production cost.2. Complex scaling: requires coordinated expansion of electrolyzers, $CO_2$ capture, and Fischer-Tropsch units.3. Infrastructure and certification: needs new airport logistics and lengthy certification processes.4. Securing long-term offtake agreements is critical for investment. | SkyNRG, Neste, LanzaTech; Lufthansa test flights. |
| E-gasoline | Fischer-Tropsch or MTG:MTG route: $2CH_3OH \rightarrow CH_3OCH_3$ (DME) + $H_2O$, then oligomerization and hydrotreating.Conditions: 190–350°C, 20–50 bar, Fe catalysts. | Drop-in fuel for existing spark-ignition (gasoline) engines in light-duty vehicles. | Low-carbon when made with renewable $H_2$. Challenges from energy-intensive synthesis and feedstock variability. | 1. Lower climate benefit per kWh than direct vehicle electrification.2. Complex, multi-stage production (methanol synthesis, MTG/Fischer-Tropsch, upgrading) .3. Scaling feedstock supply (renewable methanol, $CO_2$) is a central hurdle.4. Market challenges: regulatory classification, consumer acceptance of cost premium. | ExxonMobil/Haldor Topsoe TIGAS process; 3,000 ton/ day plant in China (2011). |

This search strategy was designed to include all relevant research articles on the topic while excluding review articles. To ensure paper quality, certain criteria were applied. Only English-language final published articles from scientific journals and conference papers were included. Sources such as book chapters, book series, and editorials were excluded because this analysis relies on final, citable research outputs that have undergone peer review, which is the standard for assessing research progress and impact in an engineering field. This article selection process shows how the dataset was filtered based on these criteria. All documents were exported from Scopus in CSV format, containing full bibliographic records, abstracts, keywords, and citation data. Duplicate entries were checked; none were found due to the use of a single database. Bibliometric analysis was conducted using VOSviewer and the Bibliometrix package in R. Bibliometrix facilitates the conversion of Scopus data exports into standardized bibliometric visualizations via the Biblioshiny interface. VOSviewer, on the other hand, generates visual representations of bibliometric networks, emphasizing relationships between keywords, authors, and citations. Key indicators analyzed included highly cited journals, authors, and papers, publication trends, and the geographical distribution of research output. Scientometric analyses, such as keyword co-occurrence and co-citation structures, revealed connections between researchers and institutions, as well as co-authorship patterns. The workflow comprised data import, descriptive statistics, network visualization, and interpretative analysis. All search parameters, exclusion criteria, and analytical settings have been explicitly documented to enable replication. The dataset, search query, and filtering steps are transparently reported to support reproducibility and future updates. Fig. 1 illustrates the flow diagram of the literature selection process.

## 4. Results and analysis

### 4.1. Key bibliometric information

The bibliometric data indicate a growing interest and activity in research published between 2003 and 2024. A total of 30 documents from 23 sources were analysed, with an annual growth rate of 3.36%. The study involved 149 authors, with only one publication by individual

authors. The 36.67% international co-authorship percentage demonstrates how widespread cross-border collaborations are in this field of study. On average, each paper had 5.43 co-authors, and all the documents cited a total of 1298 references. The analysis identified 108 authors' keywords, which demonstrate the diverse themes within membrane reactor technologies for efuel production. The average age of the documents was 6.37 years. Each document received an average of 18.33 citations. This bibliometric analysis shows the growth and international collaboration within this research area. Fig. 2 presents a summary of key bibliometric indicators derived from this study.

The growth of the publications is shown in Fig. 3. It illustrates the progression of the field from an early stage to a phase of active growth. The field experienced minimal activity in the early years, with very few publications. There was a slight increase in research output from 2016 onwards, with a noticeable uptick in 2023, when 5 articles were published. However, 2024 saw a slight decrease after that, with only 4 articles published. Despite this, the overall trend indicates a moderate interest in the subject, with an increased awareness of the importance of this research area.

The word cloud in Fig. 4 reveals key research themes focusing on membrane technology, bioreactors, and renewable energy processes. This figure visually maps the conceptual and thematic structure of the research field. It reveals how interdisciplinary themes converge around the common goal of improving energy efficiency and process intensification. Prominent terms such as membrane reactors, biogas, $CH_4$, and $CO_2$ highlight the focus on sustainable energy conversion through waste gas utilization in synthetic fuel production. This includes capturing and converting $CO_2$ and $CH_4$ to reduce greenhouse gases while improving the energy efficiency of fuel production processes. The focus is on $H_2$ production, bio-methanation, and methanol synthesis. Membrane bioreactors are used for renewable fuel and biofuel production. Energy efficiency, electrolysis, and catalysis are important for improving membrane reactors. Research focuses on improving fuel production systems and reducing energy consumption. Economic studies, such as cost-benefit and sensitivity analysis, show that assessing the feasibility of these technologies is critical for large-scale use. Transesterification for biodiesel production and reaction kinetics indicate that membrane reactor technology will improve both in performance and economics. Moving forward, research will likely contribute to improving membrane reactor performance, developing new catalysts, and improving the economic and environmental viability of these technologies for broader adoption in efuel production systems.

### 3.2. Keyword co-occurrence network

Fig. 5 presents the network visualization of author keywords using the fractional counting approach. This figure categorizes research keywords based on their centrality (relevance) and density (development) within the field of membrane reactors for efuel production. The analysis identified a total of 430 items distributed across 6 clusters, with 7271 links and an overall link strength of 304.50. This visualization depicts the relationships and connectivity between key terms in this research domain.

Among 6 clusters, cluster 1 contains 90 items. This is defined by

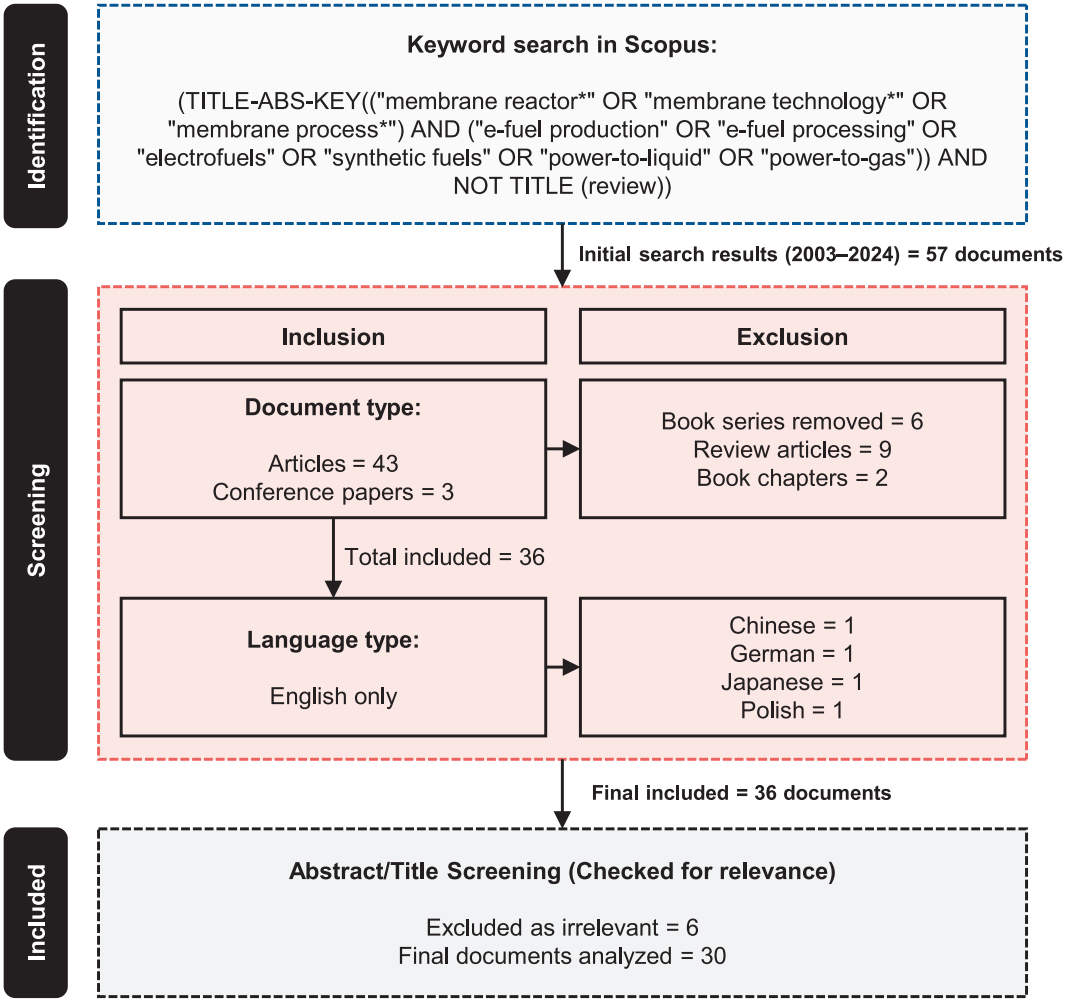



**Fig. 1.** Flowchart of the methodology adopted for the review.




**Fig. 2.** Bibliometric analysis of research publications: Document trends, author contributions, international collaboration, and citation impact.

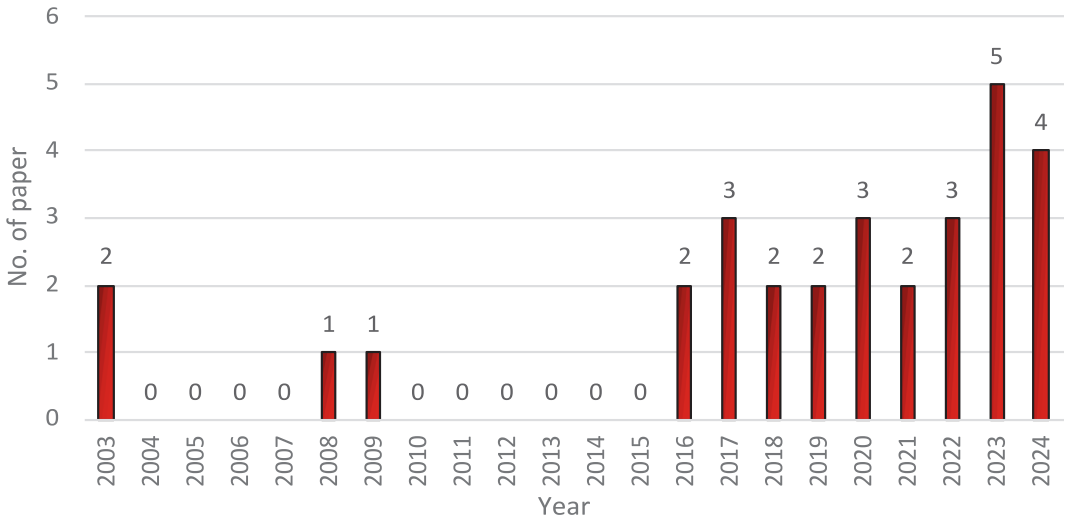



**Fig. 3.** Chronological progression of published papers over time.

keywords: methanol synthesis, $CO_2$ hydrogenation, membrane reactors, zeolite membranes, Aspen Custom Modeler, modeling and simulation, power-to-liquid, equation-oriented flowsheets, reactor performance, and synthetic fuel. These keywords are central to membrane reactors for efuel production, as they emphasize transforming $CO_2$ and $H_2$ into methanol, a key efuel, using advanced reactor designs. Methanol synthesis and $CO_2$ hydrogenation highlight the chemical processes converting $CO_2$ into liquid fuels, critical for carbon–neutral energy systems. Zeolite membranes, known for selective water removal, increase reaction efficiency by shifting the equilibrium, improving methanol yield and selectivity.

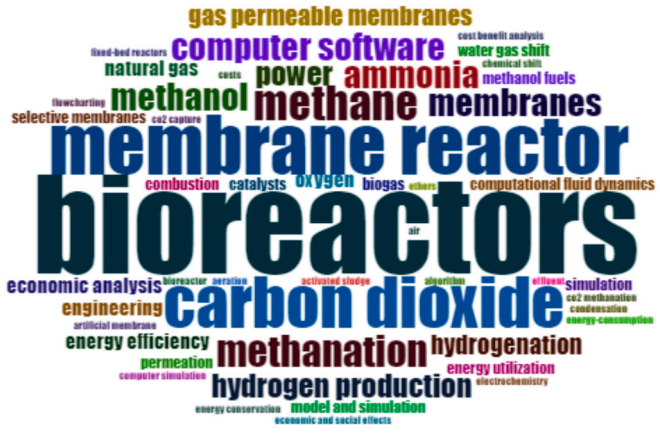



**Fig. 4.** Word cloud analysis of authors' keywords.

Aspen Custom Modeler is used for precise analysis of reactor dynamics, which optimizes operating conditions such as temperature and pressure. Power-to-liquid shows the integration of these processes into sustainable fuel production, while equation-oriented flowsheets and reactor performance metrics ensure robust system design for industrial scalability. Synthetic fuel ties these efforts to efuel production, aiming for renewable alternatives to fossil fuels. For instance, Ref. [53] developed a one-dimensional simulation framework for tubular membrane reactors applied to $CO_2$-to-methanol synthesis, implemented in Aspen Custom Modeler. The model incorporated rigorous mass and energy balances, treated forward and reverse permeation as isenthalpic processes, and explicitly modeled methanol synthesis alongside the reverse water–gas shift reaction, avoiding oversimplified assumptions. It examined feed conditions of 50 kmol/h $CO_2$, a $H_2/CO_2$ ratio of 3, inlet temperatures of 513 K, and pressures ranging from 2 to 7.5 MPa for adiabatic and nonadiabatic configurations with ideal water-selective membranes. In adiabatic reactors, increasing pressure raised conversion from 15.3% to 26.1% but reduced selectivity from 56.4% to 47.2% due to enhanced reverse water–gas shift reaction. Nonadiabatic reactors nearly doubled conversion from 21.1% to 41.4% at 7.5 Mpa, with only marginal selectivity loss. Hydrogen sweep gas facilitated water removal, limiting $H_2$ crossover, enabling adiabatic reactors with 60 membranes to achieve conversions up to 69.8% at 2 Mpa and nonadiabatic reactors to reach 55.5% with improved selectivity at low pressures. The study concluded that membrane reactors significantly improved $CO_2$-to-methanol conversion, particularly under nonadiabatic cooling and

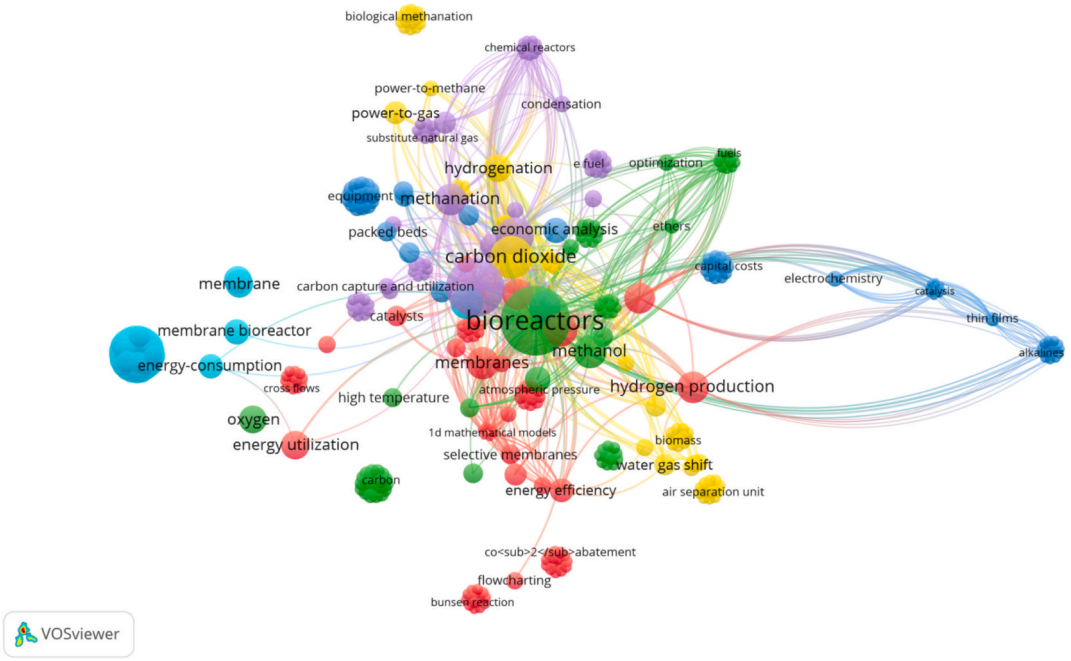



**Fig. 5.** Co-occurrence network of the authors' keywords. This network is divided into 6 clusters, including 430 keywords.

elevated pressures, with selectivity largely preserved or enhanced. The framework's versatility and adaptability supported integration into process flowsheets for sustainability assessments. In addition, Ref. [54] investigated a steam-selective membrane reactor for methanol synthesis from $CO_2$ and $H_2$, compared to a conventional reactor, using simulations over 150–310°C and 10–70 atm. High pressure and low temperature favored methanol formation, but kinetic limitations and reverse water–gas shift side reactions constrained performance. The membrane reactor used a sodalite zeolite membrane to selectively remove steam, shifting the equilibrium toward methanol, improving thermal management, and enhancing catalyst protection. A split-feed configuration reduced reactant loss at elevated temperatures by co-feeding both reactor chambers. Three performance indicators were defined: system yield, catalytic-bed yield, and global yield, with the latter incorporating permeate recycling. At 270°C and 50 atm, the membrane reactor achieved ~45% steam removal, increasing catalytic-bed methanol yield by up to 18% relative to the traditional reactor. An idealized $H_2O$-selective membrane improved yields by ~31% and permeate recycling surpassed equilibrium predictions. The study concluded that steam-selective membrane reactors, particularly with split-feed and permeate recycling, delivered substantial improvements in methanol production, though further optimization required economic assessment, improved permeation data, and multi-tube scale-up.

Cluster 2 contains 82 items, defined by keywords: dimethyl ether (DME), membrane technology, porous plate reactor, combustion, computational fluid dynamics (CFD), flow visualization, oxy-combustion, reactor engineering, and optimization. These terms are directly relevant to membrane reactors for efuel production, focusing on DME synthesis, a versatile efuel produced via $CO_2$ hydrogenation and methanol dehydration. Membrane technology enhances DME yield by selective water removal, addressing equilibrium limitations. Reactor engineering and optimization ensure efficient design and operation, critical for industrial scalability. CFD and flow visualization provide detailed insights into reactor dynamics, improving mixing and thermal management. Combustion and oxy-combustion keywords relate to the evaluation of reactor designs for fuel production, ensuring stability and efficiency in high-temperature environments, which is vital for integrating DME into decarbonized energy systems. For example, Ref. [55] investigated a decarbonization pathway for steel production by converting blast furnace $CO_2$ into DME through integrated membrane-assisted reactors, coupling $CO_2$ hydrogenation to methanol with methanol dehydration at 200–260°C and 10–20 bar. Hybrid CuO–ZnO/ZSM-5 catalysts, prepared by coprecipitation, maximized $CO_2$ conversion and methanol yield, while carbon molecular sieve membranes offered high $CO_2$ selectivity and stability in water-rich conditions. Two configurations were evaluated: membrane reactors for simultaneous $CO_2$ dosing and water removal, and membrane-assisted reactive distillation integrating catalytic conversion, separation, and distillation. Simulation results showed membrane-assisted reactive distillation enhanced DME productivity by 13% and achieved purity exceeding 99%. Multiscale analysis confirmed that catalyst morphology, membrane selectivity, reactor architecture, and process integration drove performance improvements. The study concluded that membrane-assisted $CO_2$-to-DME systems were promising for industrial decarbonization by intensifying reactions and lowering energy demand, though challenges included optimizing operating conditions, scaling up reactor configurations, and ensuring long-term membrane durability. Moreover, Ref. [56] examined oxy-methane combustion in a low-power reactor with porous alumina plates designed to emulate high-temperature membrane reactors. Combining experimental methods with CFD, the study analyzed flow characteristics, mixing behavior, and combustion performance across global equivalence ratios ($\phi$ = 0.4–1.0). Cold-flow tests using particle image velocimetry and CFD simulations confirmed laminar flow, intensified mixing near plate trailing edges, and experimental–numerical agreement with errors below 4%. Mixing improved under leaner conditions, and CFD mapping verified the imposed equivalence ratio downstream of the plates. In reacting simulations at ~3 kW, flames anchored downstream, with stoichiometric conditions producing elongated flames and lean conditions yielding shorter flames. Temperatures ranged from ~1130 K at $\phi = 0.4$ to 1240 K at $\phi = 1.0$. The reactor sustained hot zones near the plates, favoring $O_2$ permeation, while downstream flame stabilization reduced thermal stress. The study concluded that porous-plate reactors effectively emulated high-temperature membrane reactors, enhancing mixing, stabilizing flames, and protecting membrane structures, with stoichiometric operation optimal for membrane-based oxy-fuel systems.

Cluster 3 contains 74 items, defined by keywords: economic uncertainty, gases, $CO_2$ utilization, protonic, electrolytes, chromium alloys,

natural gas substitutes, quantitative analysis, and sensitivity analysis. These terms connect to membrane reactors for efuel production by emphasizing the production of synthetic natural gas and $H_2$, key efuels, through $CO_2$ utilization. Economic uncertainty and sensitivity analysis highlight the need for cost-effective solutions, critical for industrial adoption. Protonic electrolytes and chromium alloys relate to advanced cell designs for $H_2$ production, enhancing efficiency and durability. Natural gas substitutes and $CO_2$ utilization underscore the role of membrane reactors in producing renewable fuels, while quantitative analysis ensures robust evaluation of technical and economic feasibility. To illustrate, Ref. [57] evaluated synthetic natural gas production via $CO_2$ methanation in a membrane reactor with $H_2O$-permeable membranes, compared to a conventional packed-bed reactor. Combining process simulations, economic assessments, sensitivity analyses, and Monte Carlo probability modeling, the study showed that continuous $H_2O$ removal in the membrane reactor enhanced $CO_2$ conversion and $CH_4$ yield, reducing production costs by ~5% at pilot scale (50 kg/day) and ~8% at industrial scale (1000 kg/day) relative to the packed-bed reactor. Raw material costs, particularly $H_2$, and labor were dominant cost drivers, with raw materials more influential at larger scales. Monte Carlo simulations indicated economic uncertainty but showed lower cost ranges for the membrane reactor, reaching 1.01 $/kg, compared to higher costs for the packed-bed reactor. Furthermore, Ref. [58] advanced metal-supported protonic ceramic cells for steam electrolysis, achieving mechanical robustness and compatibility between Ba-based proton conductors and ferritic steel. The cell architecture reduced ceramic content by ~90% and NiO content by ~95.7%, improving scalability and reducing costs. It included a porous Fe–Cr alloy substrate with a $La_{0.80}Sr_{0.20}MnO_3 - \delta$ barrier layer, dual NiO–BaZr–Ce–Y (BZCY) electrodes, a thin BZCY721 electrolyte, and $O_2$ electrodes of platinum or Ba–Gd–La–Co–BZCY541 composite. Fabrication below 1000°C using pore-graded multilayer processes and pulsed laser deposition resulted in smooth, crack-free surfaces and compressive electrolyte stresses (~−355 Mpa). The $O_2$-evolution reaction was the primary kinetic limitation, with impedance spectroscopy indicating dominant ohmic resistance (~0.56 eV) and higher polarization resistance at the $O_2$ electrode (~1.32 eV). Degradation analysis revealed chromium and manganese diffusion up to 10 μm and partial delamination at the $La_{0.80}Sr_{0.20}MnO_3 - \delta$/Ni–BZCY721 interface. The study demonstrated that scalable metal-supported protonic ceramic cells for $H_2$ production in PtX applications, though sealing and chromium stability required further improvement.

Cluster 4 contains 67 items, defined by keywords: $CO_2$ methanation, ruthenium catalyst, power-to-methane, water–gas shift, ammonia plants, techno-economic study, industrial plants, chemical shift, and turbine inlet temperature. These terms are highly relevant to membrane reactors for efuel production, focusing on $CO_2$ methanation for synthetic $CH_4$ and ammonia synthesis, both critical efuels. Ruthenium catalysts enhance reaction kinetics, while power-to-methane and water–gas shift processes optimize $CH_4$ production. Ammonia plants and techno-economic studies address scalability and cost-effectiveness, and turbine inlet temperature ensures compatibility with energy systems. For example, Ref. [59] investigated membrane reactors for the Sabatier reaction, benchmarking against fixed-bed reactors in PtX systems. A validated non-isothermal plug-flow model showed traditional reactors achieved 74.6% conversion, while membrane reactors with sodalite membranes reached 66.1% due to $H_2$ and $CO_2$ permeation losses. Lowering temperatures improved water selectivity but slowed kinetics. An idealized $H_2O$-selective membrane achieved 80.9% conversion, and a split-feed strategy raised catalytic-bed conversion to 90.2%, though overall system conversion dropped to 54% due to ~52% $CH_4$ permeation. The study concluded that membrane selectivity was the primary limitation, but optimized membranes could surpass equilibrium performance for synthetic fuel production. In another study, Ref. [60] modeled a multi-tubular catalytic membrane reactor for renewable ammonia production in power-to-ammonia systems using Aspen Plus™ and Aspen Custom Modeler™. Assuming plug-flow behavior and ruthenium-on-carbon kinetics at 320–400°C and up to 130 bar, the study analyzed four configurations: proton exchange membrane (PEM)-traditional reactor with condensation, PEM-membrane reactor with condensation, idealized PEM-membrane reactor without condensation, and solid oxide electrolyzer cell membrane reactor with condensation. Catalytic membrane reactor improved nitrogen conversion by over 90%, though performance was limited by permeance beyond a selectivity threshold of 10–20. Techno-economic analysis showed PEM-membrane reactors with condensation increased efficiency by ~8% at a 2% cost penalty, while a solid oxide electrolyzer cell membrane reactor with condensation achieved 16% efficiency improvement but a 10% cost increase. Optimal specific energy consumptions were 9.98 MWh/t $NH_3$ for the PEM-membrane reactor with condensation and 8.45 MWh/t $NH_3$ for the solid oxide electrolyzer cell membrane reactor with condensation at ~23 t/day. Alternative purification methods were needed to maximize catalytic membrane reactor benefits.

Cluster 5 contains 64 items, defined by keywords: biogas, biological ex situ methanation, carbon capture and utilization, membrane reactor, Sabatier reaction, $CH_4$, renewable energy, reactor design, modeling and simulation, and purification. These terms directly relate to membrane reactors for efuel production by concentrating on biogas upgrading through $CO_2$ methanation to produce renewable $CH_4$. Biological ex situ methanation and membrane reactors enhance $CH_4$ purity and process efficiency, while carbon capture and utilization support sustainable energy integration. Reactor design and purification strategies minimize downstream processing, and modeling/simulation ensures optimized performance. For instance, Ref. [61] developed an inverted membrane reactor for biological methanation, using ultrafiltration hollow-fiber modules for bubble-free gas transfer from the shell side. Operated with 12 L of municipal digester sludge at ~35°C for 412 days without inoculum replacement, the inverted membrane reactor maintained $CH_4$ concentrations of 90–98 vol% with $H_2$ slip below 5 vol%. Only three nutrient additions were needed, demonstrating system stability. Maximum performance reached ~88 LN $CH_4$ $m^{-2}$ $d^{-1}$ and 2.28 LN $CH_4$ $L^{-1}$ $d^{-1}$, among the highest for long-term biological methanation. Hydraulic pressure governed $H_2$ uptake, with static mixers improving transfer efficiency by ~50%. Longer hydraulic retention times and an $H_2$ and $CO_2$ ratio near 4 ensured complete $CO_2$ conversion and stable pH. The inverted membrane reactor provided a robust, scalable platform for high-purity $CH_4$ production, supporting renewable energy integration. Also, Ref. [62] modeled two membrane reactor configurations for biogas upgrading via $CO_2$ hydrogenation, using a carbon molecular sieve membrane for $H_2$ dosing and water removal, and a palladium membrane as a selective $H_2$ distributor. A lab-scale, single-tube membrane reactor model assumed isothermal plug-flow behavior and nickel catalyst kinetics at 1 bar, 300–350°C, and $CH_4$ and $CO_2$ ratios of 1:1 to 5:1. This reactor, with $H_2$ permeance of $1.2 \times 10^{-7}$ mol $m^{-2}s^{-1}Pa^{-1}$, surpassed equilibrium conversion and reduced $H_2$ in the renewable natural gas product. The Pd-membrane reactor, with higher permeance ($2.08 \times 10^{-6}$ $molm^{-2}s^{-1}Pa^{-1}$), matched packed-bed reactor conversions but minimized $H_2$ slip, easing purification.

Finally, Cluster 6 contains one item, defined by keywords: membrane bioreactor, biological water treatment, wastewater treatment, effluents, energy consumption, control strategies, modeling simulation, water purification, and aeration control. These terms primarily focus on membrane bioreactors for wastewater treatment, emphasizing energy efficiency and process optimization. While not directly focused on efuel production, the keywords are relevant through their connection to energy-efficient membrane reactor designs and control strategies, which share principles with efuel synthesis systems. For instance, modeling and simulation optimize reactor performance, similar to their role in efuel reactors, and energy consumption reduction aligns with the sustainability goals of efuel production. Biological water treatment and water purification could support efuel processes by providing clean water for $H_2$ production or other upstream processes. Ref. [63] examined the influence of hydrodynamic conditions on the performance and

energy efficiency of suspended-catalyst photocatalytic membrane reactors for wastewater treatment. Recognizing photocatalysis as an effective advanced oxidation process, the study addressed fouling challenges associated with suspended catalysts, which offer superior mass transfer and conversion rates. An optimization framework was established, relating shear rate and transmembrane pressure to flux, pollutant conversion, and specific energy consumption. Two reactor configurations were evaluated: the oscillatory shear membrane reactor, using mechanical oscillation to generate shear, and the crossflow membrane reactor, employing pumped circulation. Experiments utilized methylene blue as a model contaminant and ZnO as the photocatalyst under ultraviolet illumination. Results indicated that fouling initially reduced flux but stabilized at steady-state levels, with increased shear mitigating resistance and enhancing degradation. The optimal ZnO concentration was 1 $gL^{-1}$. Energy analysis targeting 90% pollutant removal revealed trade-offs: in the oscillatory shear membrane reactor, specific energy consumption decreased with increasing shear until an optimum, after which oscillation power dominated, with optimal shear rates shifting lower upon scale-up. In a crossflow membrane reactor, optimal crossflow velocities also decreased with scale, but higher pumping requirements increased energy use in larger systems. The oscillatory shear membrane reactor outperformed the crossflow membrane reactor, achieving lower energy consumption and smaller reactor footprints. The study concluded that optimizing transmembrane pressure and shear was critical for balancing energy demand and treatment efficiency, with oscillatory systems showing potential, though economic feasibility, material durability, and life-cycle considerations required further exploration

Overall, the keyword analysis shows distinct research clusters with varying strengths, gaps, and maturity levels. The red cluster focuses on core membrane reactor concepts. It is highly mature and strongly connected to bioreactors, membranes, and hydrogen production. The blue cluster studies membrane design and energy efficiency. It shows moderate to high maturity but has limited links to economic or environmental aspects. Emerging clusters include yellow, which focuses on $CO_2$ conversion and methanation, and green, which studies methanol and DME synthesis. Both are growing quickly but remain less developed at industrial level. The purple cluster combines carbon capture with economic analysis, reflecting an increasing focus on commercial feasibility. The light blue cluster on electrochemistry and catalysis is specialized and still emerging, with weak links to broader efuel research. Overall, the analysis suggests that while core membrane research is mature, there is room to improve integration, scale-up, and sustainability studies.

While the clusters represent distinct thematic foci, their interconnections reveal a cohesive research ecosystem advancing membrane reactors for efuels. The core synthesis pathways in Clusters 1 (methanol) and 2 (DME) are fundamentally linked, as DME is often produced from methanol, sharing catalysts, membrane designs, and process intensification strategies like reactive distillation. These catalytic and separation principles extend to Clusters 4 (methanation/ammonia) and 5 (biogas upgrading), where similar membrane-assisted equilibrium shifting is applied to produce methane and ammonia, underscoring a unified approach to $CO_2$ valorization. The economic and scalability concerns highlighted in Cluster 3 (economic uncertainty, protonic cells) directly intersect with all technical clusters, emphasizing that advances in catalyst efficiency (Clusters 1, 2, 4) or reactor design (Cluster 5) must be evaluated against cost, durability, and system integration, themes that dominate techno-economic studies across clusters. Furthermore, Cluster 6 (wastewater treatment), though applicationally distinct, contributes cross-cutting insights into membrane fouling, hydrodynamic optimization, and energy-efficient operation, which are transferable to efuel synthesis reactors, particularly in managing water-rich streams or optimizing membrane performance under continuous operation. Thus, the clusters are not isolated but form an interdependent network where material innovation, process intensification, economic viability, and system integration continuously inform and accelerate the development of membrane reactors for sustainable efuel production.

The analysis of keyword co-occurrence networks reveals critical insights into the research landscape of membrane reactors for efuel production and their pivotal role in advancing carbon–neutral energy systems. The dominance of keywords such as methanol synthesis, $CO_2$ hydrogenation, and zeolite membranes in Cluster 1 underscores the focus on transforming $CO_2$ into methanol, a versatile efuel, with membrane reactors enhancing efficiency through selective water removal and equilibrium shifting. Studies conducted by [53] and [54] demonstrate that advanced reactor configurations, including nonadiabatic designs and steam-selective membranes, significantly improve conversion rates and selectivity, addressing kinetic and thermodynamic limitations. Cluster 2′s emphasis on DME synthesis and oxy-combustion shows the potential of membrane-assisted systems for intensifying reactions and reducing energy demands, as evidenced by [55], where integrated membrane-assisted reactive distillation achieved high DME purity. Clusters 3, 4, and 5 further illustrate the versatility of membrane reactors in producing synthetic natural gas and ammonia, with innovations such as protonic ceramic cells and biological methanation offering scalable, cost-effective solutions for $CO_2$ utilization. However, challenges persist, including membrane selectivity limitations [59] and the need for economic assessments to ensure industrial viability [57]. Cluster 6, while focused on wastewater treatment, suggests transferable principles in energy-efficient reactor design, which could inform sustainable efuel production. Collectively, these findings indicate that membrane reactors are a promising technology for efuel synthesis, but future research must prioritize optimizing membrane selectivity, scaling reactor configurations, and conducting comprehensive techno-economic analyses to bridge the gap between laboratory advancements and industrial deployment.

### 3.3. Conceptual structure: Thematic map and factorial analysis

The conceptual framework developed during the research period is depicted in a thematic map (Fig. 6), a bibliometric tool that visualizes relationships between publications and research topics by categorizing them into four quadrants based on centrality (relevance to the field) and density (maturity of a theme). Motor themes in the upper-right quadrant represent highly developed, central topics driving research progress, while niche themes in the upper-left quadrant include well-developed but isolated topics, often specialized subfields. The lower-left quadrant, encompassing emerging or declining themes, shows underdeveloped or marginal topics, indicating either emerging areas gaining traction or outdated ones losing relevance. Basic themes in the lower-right quadrant denote highly relevant but less mature topics with significant potential for future exploration. This classification reveals knowledge clusters, trends, and gaps, facilitating the identification of influential articles, prominent authors, and interdisciplinary connections, thereby guiding future research directions and collaborative efforts. A detailed analysis of the various quadrants is presented as follows:

Niche Themes, characterized by lower development density, include cost-related topics such as gas emissions, costs, and gasification, suggesting emerging or specialized research areas with limited current impact. Emerging or Declining Themes, occupying a moderate development density, encompass membrane area, power to gas, and electrochemistry, indicating areas of transitional significance that may be gaining or losing traction. Basic Themes, with higher development density, feature established topics such as computational fluid dynamics, fermentation, combustion, hydrogen production, and membrane reactors, underscoring their foundational role in the field. On the relevance spectrum, peripheral topics like cost-benefit analysis and packed beds exhibit lower centrality, while core areas such as ammonia energy utilization, $O_2$, bioreactor membrane reactors, computer software, methanol, and carbon dioxide highlight their critical influence. The intersection of these axes reveals a dynamic research network, with membrane reactors and associated technologies like hydrogen

production and electrochemistry serving as pivotal nodes. This structure suggests that membrane reactors for efuel production are a central and evolving research domain, supported by some established methodologies.

Factorial analysis is used to identify trends, condense large datasets, reveal a topic's intellectual structure, and highlight key research gaps. This method aids in understanding thematic content, detecting knowledge deficiencies, and guiding future studies. The factorial analysis of the author's keywords is presented in Fig. 7. It visualizes relationships between keywords across research articles, revealing how technologies, processes, and themes are interconnected. It is structured along two dimensions: Dimension 1 (23.14%), representing relevance or centrality, and Dimension 2 (16.37%), indicating development density or research activity. The diagram identifies a single prominent cluster, depicted as a large red polygon, which encapsulates a broad array of interconnected themes central to the field. This cluster, positioned with high centrality and moderate-to-high density, includes core topics such as $CO_2$ capture, methanol, hydrogen production, and membrane reactors. $CO_2$ capture serves as a foundational element, addressing emission reduction while providing a critical feedstock for efuel synthesis. Methanol and hydrogen production are key chemical processes. They help turn captured $CO_2$ into synthetic fuels and support the use of renewable energy. Membrane reactors, including bio- and fixed-bed types, are important for separating gases and speeding up chemical reactions. Advances in membranes and reactor design make these processes more efficient. Other supporting technologies include tools like computational fluid dynamics, MATLAB, and simulation software. These tools help optimize processes and improve performance. Energy efficiency is also a major focus, showing efforts to make systems more sustainable. The research also involves ammonia, oxygen, thin films, packed beds, zeolites, and natural gas synthesis, showing a mix of traditional and new approaches. Cost-benefit analysis and economic factors are considered to ensure practical solutions. Overall, this area of research connects basic chemical processes with advanced technologies, showing its strong influence in the field.

### 3.4. Influential articles

Table 2 shows the ten most cited papers on this topic. Studying highly cited papers is important because they highlight the most influential research and key findings that shaped the field. These papers often introduce important concepts, methods, and discoveries, giving insight into scientific and technological progress. Looking at them helps researchers understand the main ideas, research trends, and how knowledge in the field has developed over time.

The principle of optimizing existing membrane systems for energy savings is well-established. Research on a full-scale municipal membrane bioreactor demonstrated that aeration, accounting for 70–80% of energy demand, could be reduced by 15–20% using an ammonia–nitrogen-based control strategy with influent-flow correction. Field trials confirmed a 20% reduction in aeration flow and a 14% decrease in blower power, yielding annual savings of ~142 MWh with a payback under three years, though fixed-frequency blowers limited further gains [64]. This study shows that membrane-based processes can achieve significant energy savings. Smart control strategies provide a method to reduce operational costs in efuel production. Extending this optimization to efuel synthesis, thermodynamic analysis of $CO_2$ methanation established that *in situ* water removal shifts the equilibrium. While a conventional Sabatier reactor showed increased conversion with pressure but declining yield with temperature (and by-product formation like 3% CO and 63% coke at 450°C), membrane-based reactors enhanced $CH_4$ yields by up to 3.2%. Optimal water removal was identified at a fraction of 0.6, with complete $CO_2$ conversion possible at a 0.99 removal fraction, albeit with increased coke deposition [65]. This work shows why membranes are useful in methanation. It provides quantitative data on the benefits, such as improved yield, and the main trade-off, which is coke formation. It also sets benchmarks for reactor performance. Subsequent reactor design work in [66] highlighted a performance trade-off: a multitubular cascade design achieved a space–time yield of ~997, outperforming membrane reactors (~26–30) but requiring 588 tubes of 1 cm diameter. This analysis shows that membranes provide integration benefits. Conventional reactors can sometimes achieve higher core conversion. Membrane use should be justified by wider system advantages, such as simpler downstream processing or more flexible operation. Further modeling of a steam-selective hydroxy sodalite zeolite membrane reactor showed that using $N_2$ sweep gas offered marginal gains but switching to $H_2$ sweep gas increased $CO_2$ conversion by up to 57% and exceeded equilibrium $CH_4$ purity. Optimal performance occurred at a permeate-to-retentate pressure ratio of ~0.45 and a sweep-to-feed flow ratio up to 5, though higher temperatures and pressures increased $CH_4$ losses and catalyst requirements [67]. This study shows that membrane reactor performance depends heavily on operational parameters and sweep gas choice. Using $H_2$ sweep can improve performance beyond thermodynamic equilibrium. A key limitation is the lack of reliable high-temperature permeance data, which requires experimental validation.

Advancements in reactor design focus on integrating multiple functions. A double-membrane heat-exchanger reactor for converting methanol to DME, methyl formate, and $H_2$ utilized H-sodalite membranes for water removal and Pd-Ag membranes for $H_2$ separation [68]. Optimized via a genetic algorithm across 13 variables, it achieved 95.1% conversion to DME and 99.6% to methyl formate, improving yields by 10% and 38%, respectively, over a non-membrane design while reducing catalyst deactivation risk. This is an important step in process intensification. Combining multiple membrane functions with heat integration can achieve near-complete conversion in a compact,

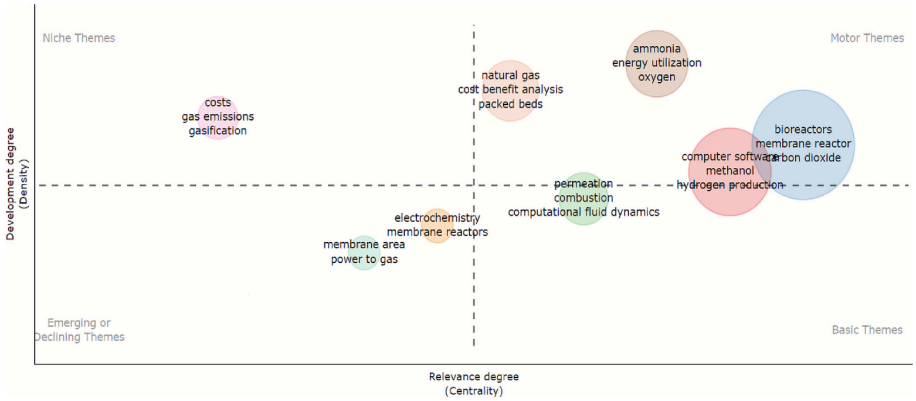



**Fig. 6.** Thematic map of the keywords used by authors.

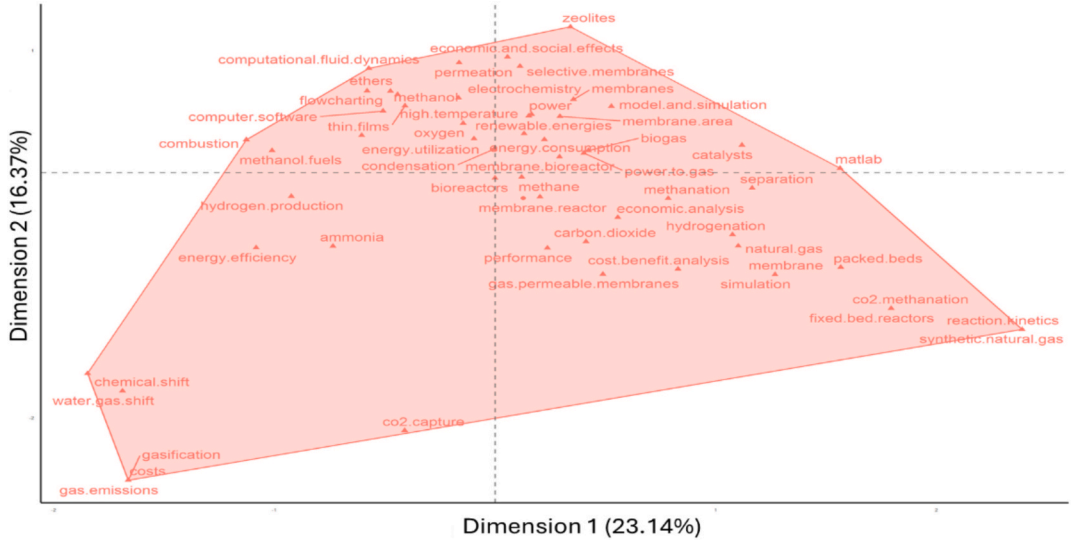



**Fig. 7.** Factorial analysis of the authors' keywords.

**Table 2**
Top ten influential papers based on citation count.

| Paper | Journal Name | Citations |
|---|---|---|
| Reducing aeration energy consumption in a large-scale membrane bioreactor: Process simulation and engineering application | Water Research | 112 |
| Biological $CO_2$-methanation: An approach to standardization | Energies | 92 |
| Thermodynamic analysis of the $CO_2$ methanation reaction with in situ water removal for biogas upgrading | Journal of $CO_2$ Utilization | 52 |
| Model-based Optimal Sabatier Reactor Design for Power-to-Gas Applications | Energy Technology | 47 |
| Development of a novel flowsheet for sulfur–iodine cycle based on the electrochemical Bunsen reaction for hydrogen production | International Journal of Hydrogen Energy | 41 |
| Optimal conditions in converting methanol to dimethyl ether, methyl formate, and hydrogen utilizing a double membrane heat exchanger reactor | Journal of Natural Gas Science and Engineering | 26 |
| Modeling and Simulation of a Steam-Selective Membrane Reactor for Enhanced $CO_2$Methanation | Industrial and Engineering Chemistry Research | 22 |
| Techno-economic comparison of power-to-ammonia and biomass-to-Ammonia plants using electrolyzer, $CO_2$ capture and water–gas-shift membrane reactor | Journal of the Taiwan Institute of Chemical Engineers | 21 |
| Integration of solid oxide fuel cell and palladium membrane reactor: Technical and economic analysis | International Journal of Hydrogen Energy | 12 |
| Integration of gas switching combustion and membrane reactors for exceeding 50% efficiency in flexible IGCC plants with near-zero $CO_2$ emissions | Energy Conversion and Management: X | 12 |

autothermal system. This approach improves efficiency and selectivity in methanol valorization. Similarly, replacing the conventional Bunsen reaction in the sulfur-iodine cycle with an electrochemical step using a PEM simplified the process, eliminating excess iodine and liquid–liquid separation. This design, producing over-azeotropic hydroiodic acid and concentrated sulfuric acid, achieved an overall thermal efficiency of 42.2% (theoretically near 50% with optimal heat recovery) via a flowsheet incorporating an $H_2$-permselective membrane reactor at 723 K [69]. This work shows that membrane reactors can be integrated into complex thermochemical cycles. They simplify flowsheets and improve energy efficiency. This makes advanced hydrogen production more practical. However, for large-scale, integrated energy systems, incorporating a membrane-assisted water–gas shift reactor into a gasification combined cycle enabled flexible operation. The base configuration achieved 47.2% net electric efficiency with 95.6% $CO_2$ capture, while an advanced design with a coal-water slurry pre-gasifier exceeded 50% efficiency and 99.5% $CO_2$ capture [70]. From this study, membrane reactors have two key roles in decarbonization. They enable very efficient fuel production and achieve near-total $CO_2$ capture. Membrane technology supports flexible, low-carbon power and hydrogen production from fossil feedstocks with carbon capture.

The transition to commercial scales highlights critical gaps in standardization and cost. Biological $CO_2$ methanation faces inconsistent reporting; a proposed framework addresses this with a three-tier matrix standardizing system description, performance metrics, microbiological reporting, and economic evaluation [71]. This work identifies a key barrier to progress: the lack of comparable data. It provides a tool to harmonize research and development, which is needed for reliable techno-economic analysis and scaling. Comparative techno-economic studies of green ammonia production pathways reveal key trade-offs. A power-to-ammonia system using alkaline electrolysis achieved ~50% efficiency, a cost of $394/ton, and a 4.95-year payback. In contrast, biomass-to-ammonia systems with water–gas-shift membrane reactors showed higher efficiency (51.6% co-current, 54.6% counter-current) but higher costs ($468/ton and $426/ton, respectively) and longer payback (6.95 and 6.35 years) [72]. This analysis gives a clear view of the efficiency-cost trade-off on a scale. Membrane reactors can improve efficiency in biomass routes, but high capital and operating costs can reduce economic viability compared to direct PtX pathways. A stark economic reality is illustrated in solid oxide fuel cell systems [73]. A palladium membrane reactor system showed higher power density (0.4398 $Wcm^{-2}$ for $CH_4$, 0.4213 $Wcm^{-2}$ for biogas) than a conventional reformer system (0.423 and 0.399 W $cm^{-2}$). However, this required high retentate pressures (29 bar for $CH_4$, 51 bar for biogas), leading to significant compressor costs and resulting in net financial losses (~$13,883 for $CH_4$, ~$32,071 for biogas), rendering it non-viable despite technical superiority. This is a landmark case study demonstrating the "valley of death" between technical success and commercial adoption. It shows that superior technical performance (higher power density) is meaningless if the system economics are negative due to high-cost membranes and parasitic energy loads.

The literature review of top-cited studies demonstrated some critical aspects: membrane reactors can significantly enhance conversion rates,

selectivity, and energy integration for efuel synthesis, from methane and methanol to ammonia and hydrogen. Success factors include optimal water removal, strategic sweep gas use, and coupled reaction-separation designs. However, the analysis consistently identifies three core limitations that curtail industrial viability: 1) Membrane durability and performance under realistic high-temperature/pressure conditions, 2) High ancillary energy demands (e.g., for compression) that erode system efficiency, and 3) Prohibitive costs, especially from precious-metal membranes and balance-of-plant components, which undermine economic feasibility even when technical performance is superior. Therefore, future progress hinges not merely on improving conversion metrics but on developing durable, low-cost membranes and conducting integrated system-level optimizations that prioritize total lifecycle cost and energy efficiency.

### 3.5. Influential authors and sources

Table 3 summarizes the contributions of various authors, detailing the number of articles each published, and the proportion of each article's content attributed to them through fractionalized values, which reflect the extent of their authorship in collaborative or sole-authored works.

L. M. Madeira and C.V. Miguel each published 4 articles, with a fractionalized contribution of 1.17, which shows they played a significant but shared role in their articles. F. Gallucci published 3 articles, with a fractionalized value of 0.68, which suggests a moderate level of responsibility in his collaborative works. S. Assabumrungrat, C. Faria, A. M. Dayaghi, I. G.I. Iwakiri, T. Norby, and S. Richard each contributed to 2 articles, with fractionalized values of 0.29, 0.58, 0.24, 0.58, 0.24, and 0.48, respectively, which demonstrate varied levels of involvement in their co-authored articles. A. A. Adesina contributed to 1 article, with a fractionalized value of 0.13, which indicates a minor role in a collaborative work. This table provides a clear overview of the authors' contributions and the diversity of their roles in the published articles.

Fig. 8 presents the distribution of articles from the top 10 sources based on publication counts, with the International Journal of Hydrogen Energy leading as the most prominent publication venue with 5 articles. This concentration indicates that hydrogen-centric research is a dominant and mature subfield, while the diversity of other high-impact journals reflects the interdisciplinary nature of efuel studies, spanning chemical engineering, sustainability, catalysis, and energy systems. Chemical Engineering and Processing – Process Intensification and Energy contributed 3 and 2 articles, respectively, with the remaining sources, such as ACS Sustainable Chemistry and Engineering, Advanced Sustainable Systems, and Catalysts, each contributing one article and adding diverse insights.

### 3.7. Geographical distribution and collaboration

Fig. 9 illustrates the global distribution of scientific output, showing the frequency of contributions from authors from various countries in this field. The network of international collaborations shown by the red lines depicts the globally interconnected and cooperative nature of this research area. Authors from Germany top the list with 45 publications, demonstrating their leading role. China follows with 23 publications that highlight its growing influence in sustainable energy research. Authors from Portugal contribute 14 publications, while Thailand adds 12, both showing notable engagement. The Netherlands, South Korea, and Spain each provide 9 publications, reinforcing their presence. Norway and the USA each contribute 7 publications, maintaining steady output. Japan adds 5 publications, while France, Italy, Poland, and Saudi Arabia each supply 4 publications, indicating diverse contributions. Iran contributes 3 publications, and Switzerland offers 2. Australia, Belgium, Canada, and Malaysia each provide 1 publication, rounding out the global participation. This distribution demonstrates a broad international effort, with Germany and China at the forefront, supported by consistent contributions from Europe, Asia, and beyond.

**Table 3**
Top ten influential authors based on paper count.

| Authors | Papers | Articles fractionalized |
|---|---|---|
| Madeira, Luis M. | 4 | 1.17 |
| Miguel, C.V. | 4 | 1.17 |
| Gallucci, Fausto | 3 | 0.68 |
| Assabumrungrat, S. | 2 | 0.29 |
| Catarina Faria, A. | 2 | 0.58 |
| Dayaghi, Amir Masoud | 2 | 0.24 |
| Iwakiri, Igor G.I. | 2 | 0.58 |
| Norby, Truls | 2 | 0.24 |
| Richard, Simon | 2 | 0.48 |
| Adesina, A.A. | 1 | 0.13 |

The country collaboration map from Fig. 9 provides an overview of international research collaborations. The data reveals a varied network of country-level interactions. Germany collaborates with Belgium, Japan, the Netherlands, Norway, and Switzerland, each partnership producing 1 paper. The Netherlands partners with Belgium and France, with France yielding 2 papers and Belgium 1. Norway collaborates with Italy and Japan, each contributing 1 paper, and with Spain, resulting in 2 papers. Poland collaborates with Iran for 1 paper, while Spain also partners with Italy for 1 paper. Thailand connects with Australia and Malaysia, each yielding 1 paper. The USA collaborates with Iran, producing 1 paper. This active international engagement highlights a broad distribution of collaborations, with frequencies ranging from 1 to 2 papers per partnership. The map underscores the emerging importance of these global networks in advancing membrane reactor technologies for efuel production, with notable ties across Europe, Asia, and North America.

## 4. Transitioning conventional fuels to sustainable efuels: A sector-specific analysis

### 4.1. Analysis of notable global projects

Table A2 presents an overview of global efuel projects and their contributions. From this table, it can be identified that these projects span diverse applications, revealing five primary sectors: aviation, maritime shipping, road transportation, steel and heavy industry, and chemicals and refining. Projects such as Norsk efuel and Fjord PtX promote e-kerosene as an alternative for aviation. It can work with current engines, avoiding expensive redesigns, but manufacturers may resist without clear standards, which could fragment the market. Local production, as in Norway, reduces dependence on oil-producing countries, though oil-rich nations might respond with trade restrictions or price cuts. EU regulations, such as ReFuelEU Aviation, encourage adoption and investment. However, high production costs may strain airlines, especially budget carriers, leading to higher ticket prices and limited accessibility. Renewable infrastructure requires land and water, which could create local conflicts, potentially delaying projects.

Heavy fuel oil and marine diesel oil are essential for long voyages. Projects like Liquid Wind and eMethanol Kassø promote e-methanol and e-ammonia as alternatives. Offshore production, like $H_2$Mare, aims to strengthen supply chains, but past cost overruns make it uncertain. Dual-fuel engines offer a retrofit solution, but shipowners may resist due to schedule or warranty concerns. E-ammonia's toxicity requires new safety regulations and increases insurance costs. Higher production costs, lower energy density, and renewable variability could reduce profitability and lead operators to continue using conventional fuels. The lack of global standards for toxic fuels further complicates adoption.

Gasoline and diesel remain dominant. Projects like Repsol Bilbao and Haru Oni promote e-diesel, which can blend with conventional fuel, using existing infrastructure. Local production boosts energy independence, but reliance on one region is risky. EU regulations may push

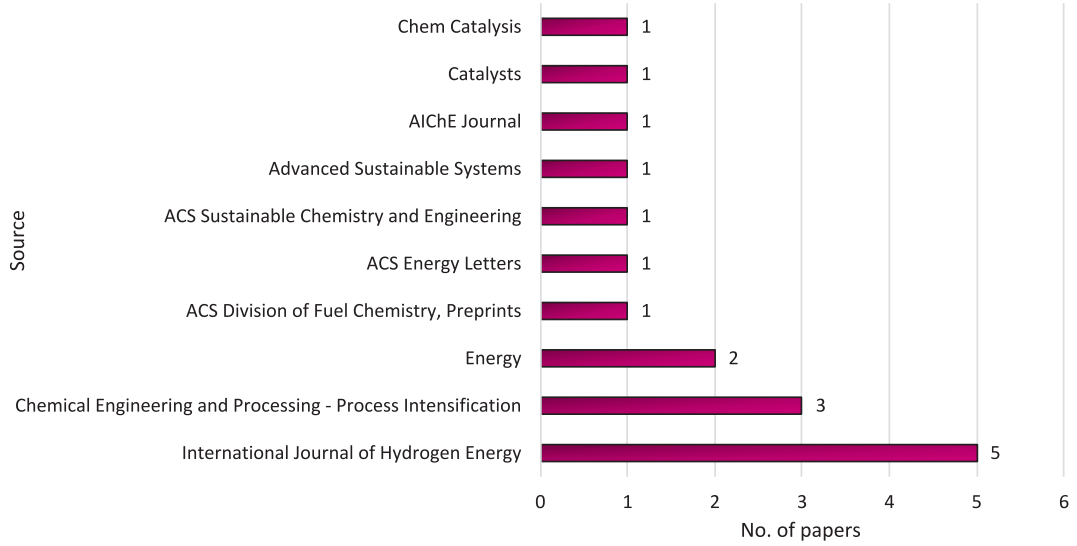



**Fig. 8.** Top ten influential journals based on paper count.

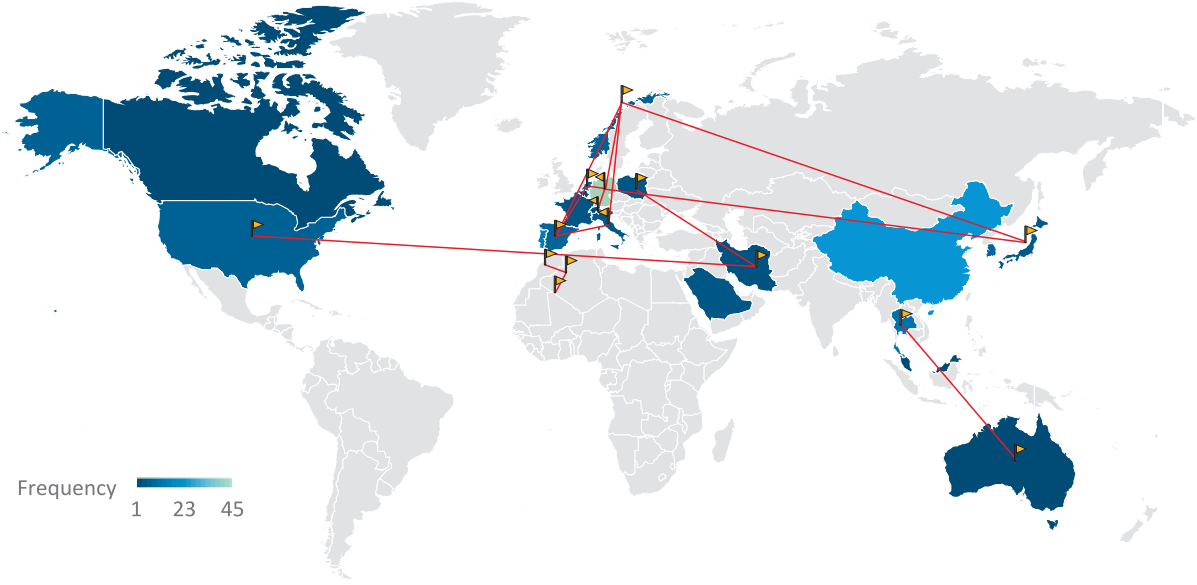



**Fig. 9.** Geographical distribution of author contributions and international research collaborations: Frequency of author contributions by country, with red lines representing the intensity of cross-border research partnerships. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

adoption, but smaller companies may struggle with costs. Higher e-diesel prices increase logistics expenses, and renewable energy projects may compete with agriculture, raising food prices. Battery electric vehicles present a competing technology, slowing e-diesel adoption. A wide range of stakeholders makes creating a unified approach challenging.

Coal, coke, and natural gas dominate production. Projects like Hybrit and Stegra $H_2$ Green Steel offer green hydrogen, but scaling is unproven, with potential supply shortages. EU funding may help, but reliance on subsidies could reduce private investment. Hydrogen production demands may strain power grids, and catalyst degradation increases maintenance costs. Upgrading facilities is expensive, risking bankruptcy for smaller producers and market consolidation. The sector's transition depends on overcoming economic, structural, and workforce challenges.

The sector relies on syngas, making it vulnerable to supply disruptions. Companies like Carbon Recycling International and Westküste 100 suggest e-methanol as a local alternative, but downstream industries must accept it. Low production efficiency and high costs may require subsidies to compete with traditional syngas. Local resource use may cause community opposition, and high electricity demand could worsen inequalities. Uncertainty and weak demand may slow investment. Overall, the transition depends on addressing economic, social, and operational challenges.

### 4.2. *Some positive and negative dynamics*

Using existing infrastructure is a big advantage. It helps industries avoid costly upgrades and reassures cautious stakeholders. Local production of efuels also boosts energy independence, reducing reliance on imports and giving regions with renewable resources more control. Policy support adds another layer of encouragement. Regulations and incentives make it easier for companies to adopt new standards. Overall, these factors create a positive environment for the transition.

- Using existing aviation engine designs avoids expensive redesigns, benefiting airlines by lowering initial investment.
- Local production in maritime projects strengthens supply chain resilience, reducing reliance on unstable oil regions.
- EU regulatory support for road transportation, such as e-diesel mandates, motivates fleet operators to embrace new fuels.
- Underground hydrogen storage in steel projects guarantees a stable supply for industrial use, ensuring operational continuity.
- Local e-methanol production in chemicals projects reduces reliance on imports.

Still, some challenges come with efuels. Higher production costs could put pressure on the market and affect both businesses and consumers. Competition for land and water might cause local conflicts and increase project expenses. New technologies bring uncertainty because untested systems could fail and slow adoption. At the same time, multiple competing solutions could divide investment and leave efuels without a clear position. These problems show that the shift involves real economic, social, and technical hurdles. More negative dynamics include:

- Higher e-kerosene costs in aviation may increase ticket prices, putting pressure on budget carriers and limiting passenger access.
- The toxicity of e-ammonia in maritime projects requires new safety measures, raising insurance and training costs for shipowners.
- Competition for land needed for renewable energy could conflict with agricultural use, potentially driving up local food prices.
- The energy demands from hydrogen could overwhelm power grids, requiring costly upgrades that smaller plants can't afford.
- Low e-methanol synthesis efficiency in chemicals could raise production costs, making it less competitive without subsidies.

These factors must be carefully managed to prevent stagnation. The balance between positive and negative dynamics will shape the future of efuels.

### *4.3. A brief implication*

The transition to efuels will happen slowly if challenges are handled well. Stakeholders must manage economic pressures, technological risks, and uncertainties in the current system. Concentrating production in certain regions could create the same power imbalances seen in oil markets. Relying too much on subsidies might make the sector depend on public support instead of the market. Local resource conflicts and a fragmented market show that the path forward will be difficult and require strong coordination among many stakeholders. The connection to industrial $CO_2$ raises a key question: are efuels a real solution or just a temporary fix? Companies and policymakers need to weigh costs against long-term goals. The future of the sector depends on creating smart governance and reliable technologies to overcome obstacles. Without progress, efuels could turn into an expensive experiment with little effect. A practical approach is needed to navigate this complex landscape and plan for realistic outcomes. Several actionable steps can be taken, including:

- Develop sector-specific regulatory frameworks to standardize efuel specifications. This ensures that engine manufacturers in aviation and shipowners in maritime shipping follow consistent safety and performance criteria.
- Establish public–private partnerships to fund infrastructure upgrades, such as grid enhancements for steel plants and water management systems for road transportation projects.
- Implement community engagement programs in regions impacted by land use, particularly agricultural areas near renewable energy sites for road and aviation projects. Transparent negotiations can address disputes and secure local support.
- Create training initiatives for maritime crews and chemical plant staff to handle toxic substances like e-ammonia and optimize e-methanol production.
- Launch pilot programs to test efuel technologies across all sectors. This will allow stakeholders to identify and resolve technical issues before full-scale implementation.
- Coordinate international forums to align subsidy policies and investment strategies. This will prevent market distortion in the chemicals sector and ensure fair support for efuel adoption across different regions.

## 5. Discussions

Using membrane reactors in PtX systems for efuel production is an important step toward cleaner energy. However, this change brings technical, economic, and operational challenges. Membranes, especially those made from precious metals like palladium, are essential for producing high-purity hydrogen. They can suffer from thermal stress, chemical damage, and fouling, which shortens their lifespan and raises costs. Adding other metals can reduce hydrogen embrittlement, but it often lowers the membranes' ability to separate hydrogen effectively. This creates a trade-off between durability and efficiency. Ceramic membranes cost less, but they have trouble keeping hydrogen very pure, which limits their use in some efuel processes. Producing efuels like ammonia and hydrocarbons requires a lot of energy. This high energy use can make efuels less environmentally friendly than direct electrification. Membrane reactors help increase conversion rates, but keeping the right pressure for separation and controlling heat in smaller systems uses a lot of energy, which reduces efficiency. Integration with fluctuating renewable energy complicates reactor control, directly affecting critical operational variables. Variations in $H_2$ feed rate, inlet pressure, and temperature disrupt stoichiometric ratios (e.g., $H_2/CO_2$), shifting reaction equilibria and conversion efficiency. Exothermic syntheses amplify these fluctuations into unstable thermal profiles and hot-spot formation, requiring precise regulation of coolant flows. Simultaneously, the permeation driving force across the membrane becomes unstable, altering separation performance and product purity. This tight coupling of feed variability, thermal inertia, and separation dynamics makes consistent reactor performance under renewable intermittency a significant control challenge. Financial issues also make adopting efuels difficult. High upfront costs for materials, complex reactor designs, and the expense of producing green hydrogen and capturing $CO_2$ are major barriers. Large investments and retrofitting old plants remain challenges, especially since clear performance standards and regulations are not yet in place. In addition, integrating membrane reactors with renewable energy systems is complicated due to the fluctuating energy inputs. This variation makes it hard to control the process efficiently, and maintaining consistent performance is still a work in progress, with advanced monitoring and control strategies still under development.

Despite these challenges, membrane reactors have strong potential for efuel production. Hybrid composites that mix metal and ceramic materials are being developed to improve stability while keeping high hydrogen purity. This approach could reduce the need for expensive precious metals and lower production costs. Silica-based and zeolite membranes are also being studied to perform better at high temperatures, making e-methane and e-methanol production more efficient. New reactor designs combine reaction, separation, and heat exchange to save energy and increase output. Modular reactors for decentralized applications, such as maritime or off-grid systems, provide practical ways to produce efuel locally without major changes to infrastructure.

From an economic point of view, membrane reactors could lower operating costs and improve efficiency. Future developments, like self-healing membranes, may extend their lifespan and reduce maintenance expenses. Technologies such as artificial intelligence and machine learning could help optimize reactor conditions in real time, boosting output and cutting energy use. Modular, plug-and-play systems could

make decentralized production easier, reducing transportation costs and improving energy security. Pairing these reactors with advanced electrolysis technologies, such as high-efficiency solid oxide electrolyzers, could lower hydrogen production costs and make efuels more affordable. Bio-inspired catalysts may also reduce energy needs in the synthesis process. Policy support, like carbon pricing or renewable fuel subsidies, can further help adoption. International research collaboration could bring innovations and create standardized guidelines, offering a path to carbon–neutral fuels in hard-to-decarbonized industries.

This study has several limitations that merit consideration. First, non-English and non-peer-reviewed sources were excluded. While this may slightly reduce global representation, only four identified articles were published in non-English languages, suggesting that most high-impact research is already accessible in English. Second, Scopus was used as the sole database. Although this choice may omit studies indexed elsewhere, Scopus offers broad multidisciplinary coverage and includes a substantial portion of high-quality peer-reviewed publications. Nevertheless, certain regional studies or alternative methodological approaches may be underrepresented. Third, the use of Boolean operators enhances search precision but may not capture all variations in terminology across disciplines, meaning that some relevant studies could have been overlooked. To mitigate this, a well-defined search strategy and systematic filtering were implemented to remove duplicates and irrelevant records, thereby improving dataset quality and reliability. Finally, the analysis identified a relatively low number of publications (30 documents), which reflects the niche and emerging character of membrane reactor research for efuel synthesis. As the field is still in its developmental stage, the findings should be interpreted with caution. Future scientometric analyses would benefit from larger datasets or longitudinal studies as publication activity increases and the field matures.

## 6. Conclusions

Membrane reactors offer a new way to produce efuels. They can improve efficiency, save energy, and help achieve global decarbonization goals. This study looks at how membrane reactors make efuels and analyzes Scopus publications from 2003 to 2024. It highlights advances in producing e-methane, e-methanol, e-ammonia, e-diesel, e-kerosene, and e-gasoline. Research focuses include methanol synthesis, DME production, $CO_2$ methanation, ammonia synthesis, and upgrading biogas. Selective membranes and reactor designs boost efficiency by reaction-separation integration. However, durability issues, high energy needs, costly materials, and complex operations limit industrial use. In conclusion, membrane reactors transform efuel production within PtX systems to aid decarbonization. They enable versatile energy carriers and chemical raw materials, utilize existing infrastructure, and suit energy storage and transportation, and industry decarbonization. From a practical perspective, success relies on affordable, durable membrane development, reactor design enhancement, and cheap, renewable energy use. Future research must address these challenges to drive a scalable, sustainable energy transition and achieve a carbon–neutral future.

## CRediT authorship contribution statement

**Rahbaar Yeassin:** Writing – original draft, Validation, Methodology, Investigation, Data curation, Conceptualization. **Prangon Chowdhury:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Prithibi Das:** Validation, Software, Resources, Data curation, Conceptualization. **Ephraim Bonah Agyekum:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Omar Farrok:** Writing – review & editing, Supervision, Visualization, Conceptualization, Methodology, Validation. **Pankaj Kumar:** Writing – review & editing, Writing – original draft, Validation, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix

**Table A1**
Comparison of gravimetric and volumetric energy densities and typical storage conditions for selected energy carriers, including hydrogen in various states [74–77].

| Energy carrier | Gravimetric energy density (MJ/kg) | Volumetric energy density (MJ/L) | Storage Conditions |
|---|---|---|---|
| Hydrogen (gas) | 120–142 | ~0.0107 | 1 bar, 20°C |
| Hydrogen (liquid) | 120–142 | ~8.5–10 | −253°C, cryogenic |
| Hydrogen (700 bar) | 120–142 | ~5.6 | 700 bar, 20°C |
| Gasoline | 44–46 | ~32 | Liquid, ambient |
| Diesel | 45–46 | ~38 | Liquid, ambient |
| Methanol | 19.9–22.7 | ~15.8 | Liquid, ambient |
| Ammonia | 18.6–22.5 | ~12.7 | Liquid, 10 bar or − 33°C |

**Table A2**
Overview of major efuel projects and their contributions.

| Project | Efuel/PtX product(s) | Primary sector | Scale/capacity | Status | Key challenge/notable feature | Ref. |
|---|---|---|---|---|---|---|
| HIF Haru Oni (2022) | e-Gasoline; e-LPG; intermediate e-methanol | Transport fuels (road, marine, aviation supply chains) | Demonstration / industrial-scale integrated efuels facility | Operating | First integrated operating efuels facility; strong dependency on renewable electricity, $CO_2$ sourcing, and permitting stability | [78] |
| Norsk efuel – Mosjøen (2020) | e-Crude with upgrading to SAF | Aviation | 50 million L/y e-Crude, up to 80% upgraded to SAF | Basic engineering/ development | Financing, electrolyzer scale-up, and long-term offtake contracting | [79] |

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**Table A2** (*continued*)

| Project | Efuel/PtX product(s) | Primary sector | Scale/capacity | Status | Key challenge/notable feature | Ref. |
|---|---|---|---|---|---|---|
| Carbon Recycling International – George Olah Plant (2011–2012) | Renewable / e-methanol ($CO_2$ hydrogenation) | Fuels and chemicals | 4,000 t/y renewable methanol | Operating | Early industrial proof; economics depend on electricity price and $CO_2/H_2$ availability | [80] |
| Sunfire co-electrolysis (demonstration; 2018–2019) | Syngas ($CO + H_2$) as PtX intermediate | PtX technology platform | High-temperature co-electrolysis test operation (>500 h) | Demonstrated at pilot scale; scaling in progress | High-temperature durability, stack lifetime, and system integration constraints | [81] |
| Hydrogen Park South Australia (HyP SA) (2021) | Renewable $H_2$ blending into the gas network | Gas grid/heat | ~1.25 MW PEM electrolyzer | Operating (hydrogen blending to customers) | Regulatory acceptance for blending; limited relevance to liquid efuels except as upstream $H_2$ learning | [82] |
| HYBRIT (2016) | Hydrogen-based steel route | Steel | Pilot/demonstration chain | Operating pilots; scale-up pathway continues | Hydrogen supply, storage, and cost; relevance is indirect through $H_2$ infrastructure learning | [83] |
| REFHYNE I (2018) | Renewable $H_2$ for refinery integration | Refining / industrial hydrogen | 10 MW PEM electrolyzer, started up in 2021 | Operating | Integration into industrial demand, electricity cost and availability | [84] |
| $H_2$FUTURE (2017) | Renewable $H_2$ | Industrial hydrogen/steel | 6 MW PEM, commissioned 2019 | Operating pilot facility | Grid-service operation under variable load; scale-up economics | [85] |
| North$H_2$ (2020) | Renewable $H_2$ (enabler for efuels) | Industry/ hydrogen backbone | Large-scale feasibility programme | Development/ feasibility | Permitting, grid/offshore wind integration, and hydrogen transport infrastructure | [86] |
| Green Fuels for Denmark (2020) | e-Methanol; e-kerosene (and renewable hydrogen) | Heavy transport (marine, aviation, road) | ~1.3 GW electrolyzer capacity by 2030 | Development (project funding and development work reported) | Long-term offtake, policy certainty, and cost reduction for RFNBO fuels | [87] |
| WESTKÜSTE100 (2020) | Renewable $H_2$, pathway includes climate-friendly fuels (PtX) | Industry + aviation fuels pathway | Demonstration / "real-world laboratory" framework | Demonstration under development/ implementation phases | Multi-partner integration, permitting, and coupling offshore wind to industrial demand | [88] |
| $H_2$Mare (2021) | Offshore $H_2$ and offshore PtX products (includes methanol/ ammonia concepts) | Offshore PtX research and development | Flagship programme | Active research and development flagship | Offshore system integration; safety; logistics; cost | [89] |
| St1 Power-to-Methanol Lappeenranta (Finland) (planning; public reporting 2022) | Synthetic/renewable methanol | Marine and road fuels | First Finnish synthetic methanol plant next to Finnsementti | Planning / pre-design reported | Project finance and electricity/ $CO_2$ sourcing; permitting | [90] |
| P2X Solutions – Harjavalta (2021) | Renewable $H_2$ (enabler for efuels) | Industrial hydrogen | Commercial-scale project development | Under construction/ development | Grid connection; equipment delivery; demand aggregation | [91] |
| Liquid Wind – Örnsköldsvik e-methanol (2017 company; project development ongoing) | e-Methanol | Marine fuel | 100,000 t/y e-methanol at Örnsköldsvik after project restructuring | Development | Offtake contracts and cost competitiveness against fossil methanol | [92] |
| Kassø e-methanol (Denmark) | e-Methanol | Marine fuel + industrial feedstock | 42,000 t/y e-methanol | Opened/operating at commercial scale | Scale-up of renewable $H_2$ and biogenic $CO_2$ integration; price premium vs fossil methanol | [93] |
| HySynGas – Brunsbüttel | Synthetic methane / e-methane (PtG) | Gas grid/industry | Tens of MW-scale | Development | Methanation economics; $CO_2$ supply; policy support for e-methane | [94] |
| HyNet North West (UK) | Low-carbon hydrogen + $CO_2$ transport and storage (enabler for efuels) | Industry + heavy transport | Programme-level infrastructure build out | Development / phased delivery | Coordinated delivery of hydrogen and $CO_2$ networks; regulatory model | [95] |
| Repsol – Bilbao synthetic fuels demo | Synthetic fuels (demo) | Refining/ transport fuels | Bilbao-area synthetic fuels demonstration plant | Development/ execution phase | Cost, hydrogen supply, and policy support for synthetic fuels in refineries | [96] |
| Power-to-Methanol Antwerp (Belgium) | e-Methanol | Marine fuel + chemicals | Project cancellation reported | Cancelled | Capital expenditure escalation and commercial risk | [97] |
| North-C-Methanol (North Sea Port) | Renewable/green methanol | Marine fuel + chemicals | ~44,000–45,000 t/y methanol with ~65 MW electrolyzer | Development | Industrial symbiosis and $CO_2$ sourcing; phased expansion | [98] |
| SolWinHy Cádiz (Spain) | e-Methanol | Marine fuel + industrial offtake | 29,000 t/y e-methanol, production start targeted early 2027 | Development | Offtake certainty and integration with dedicated renewables | [99] |
| HØST PtX (Denmark) | PtX fuels (product mix varies) | Heavy transport | Large PtX build out at Esbjerg | Development | Grid/renewables integration, and offtake contracting | [100] |
| Fjord PtX (Denmark) | efuels for aviation (PtX) | Aviation | SAF production at Aalborg with a long development horizon | Development | Commercialization timeline and policy support | [101] |
| HIF Tasmania (ongoing development) | e-Methanol (primary) | Marine fuel + export fuels | >200,000 t/y e-methanol concept | Pre-FID/permitting and design work reported | Permitting and community acceptance; $CO_2$ and renewable power integration | [78] |
| Arcadia eFuels – Project Arc (Texas, USA) | e-SAF | Aviation | Start target around 2028 | Development | Bankable offtakes and policy alignment for e-SAF | [102] |
| Arcadia eFuels – Project ENDOR (Denmark) | e-SAF (and related efuel outputs) | Aviation | FEED completion milestone | Development toward FID | Permitting, $CO_2$ logistics, and offtake | [103] |

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**Table A2** (*continued*)

| Project | Efuel/PtX product(s) | Primary sector | Scale/capacity | Status | Key challenge/notable feature | Ref. |
|---|---|---|---|---|---|---|
| INERATEC – ERA ONE (Frankfurt-Höchst) (2025) | efuels (PtL) | Aviation and transport fuels | Up to 2,500 t/y efuels | Operating | Catalyst/process integration and supply chain scale-up beyond the first commercial plant | [104] |
| Infinium – Project Roadrunner (Texas) (publicly announced 2024–2025) | e-SAF (and other efuels) | Aviation | 23,000 t/y e-SAF | Under construction | Construction execution, continuous operations, and $CO_2$ supply contracting | [105] |

## Data availability

Data will be made available on request.

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